

EE104 Fundamentals of Electric Circuits (Lesson 02)

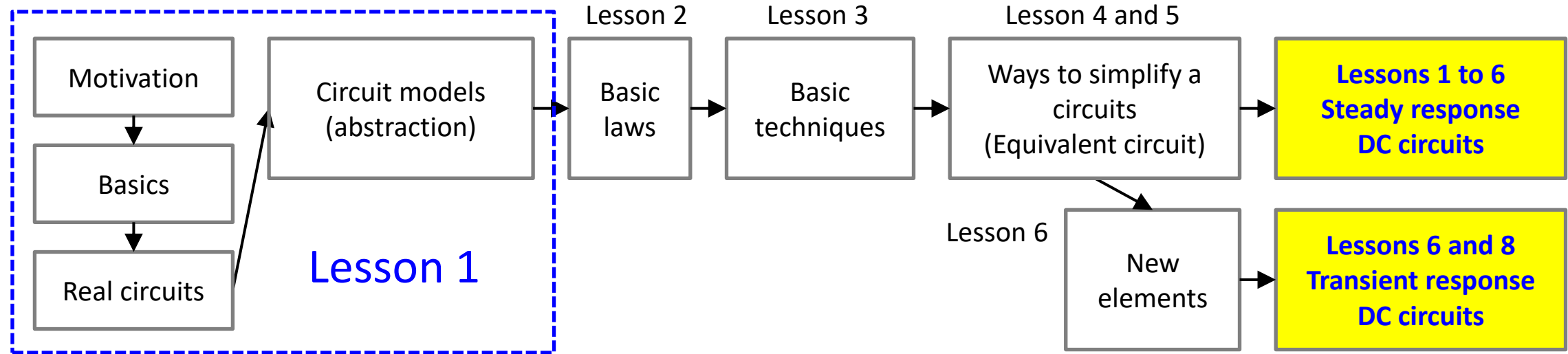
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Review of Lesson 1 (a)

- EE104 roadmap (DC portion)



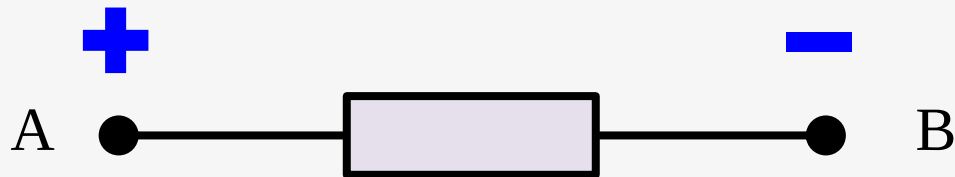
- a) **Electric circuits:** a circuit is an interconnection of electrical elements, which is necessary for almost every application that requires electricity to work.
- b) **Basic quantities:** current (A), voltage (V), power (W). To analyze a circuit is to find the current through its elements and the voltage across these elements.
- c) **Basic elements:** resistors, electric wires, dependent sources, independent sources.
- d) **Circuit models:** circuits based on lumped circuit abstraction.

Review of Lesson 1 (b)

- Reference directions

Associated reference directions

Reference voltage direction



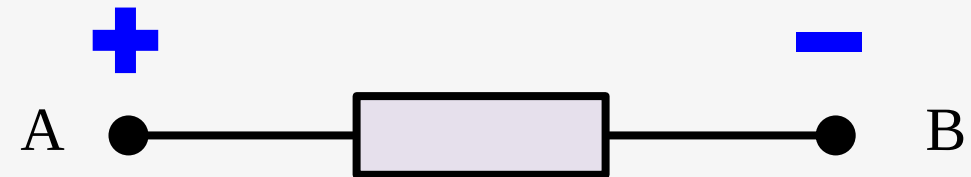
Reference current direction

$P = vi > 0$ absorb energy

$P = vi < 0$ supply energy

Nonassociated reference directions

Reference voltage direction



Reference current direction

$P = vi > 0$ supply energy

$P = vi < 0$ absorb energy

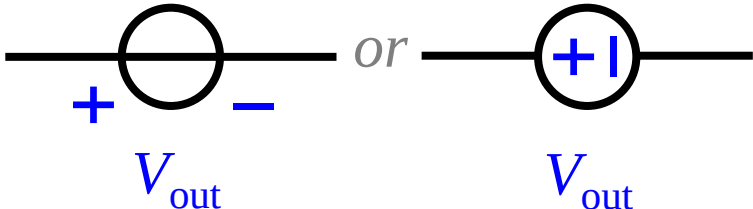
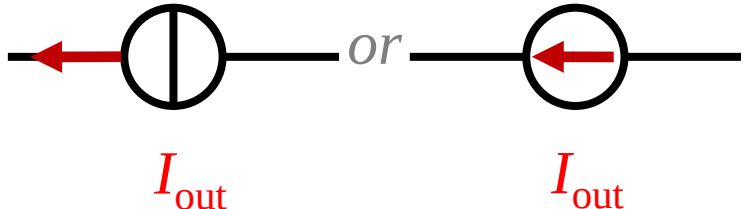
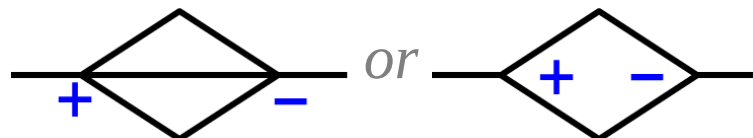
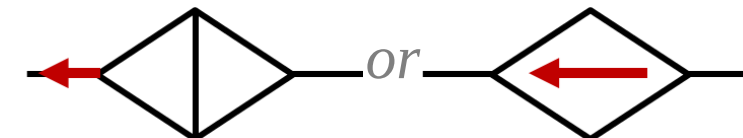
Review of Lesson 1 (c)

- **Passive Sign Convention:**

- a) In electrical engineering, the passive sign convention (PSC) is a sign convention or arbitrary standard rule adopted universally by the electrical engineering community for defining the sign of electric power in an electric circuit.
- b) The convention defines electric power flowing out of the circuit into an electrical component as positive, and power flowing into the circuit out of a component as negative. So a passive component which consumes power will have positive power dissipation, while an active component supplying power will have negative power dissipation.
- c) This is the standard definition of power in electric circuits; it is used for example in computer circuit simulation programs such as SPICE.

Review of Lesson 1 (d)

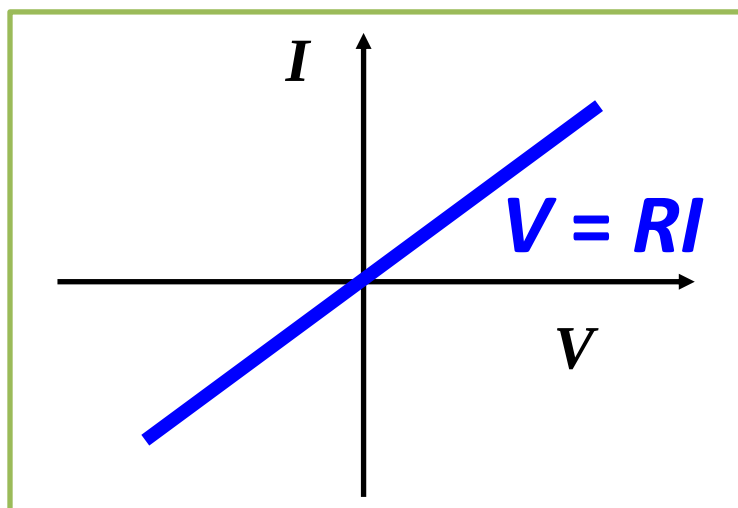
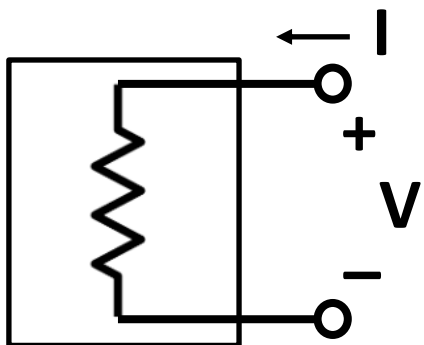
- Dependent and independent sources:

	Voltage sources	Current sources
Independent	 V_{out}	 I_{out}
Dependent	 $V_{\text{out}} = kV_{\text{in}}$ (or kI_{in})	 $I_{\text{out}} = kI_{\text{in}}$ (or kV_{in})

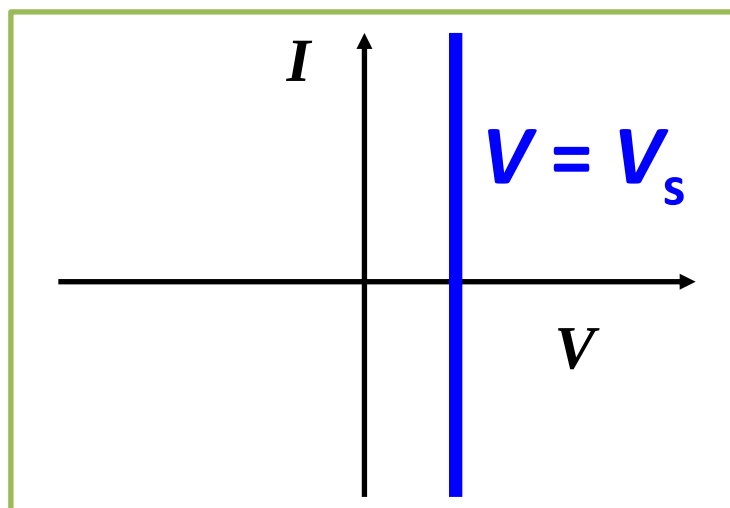
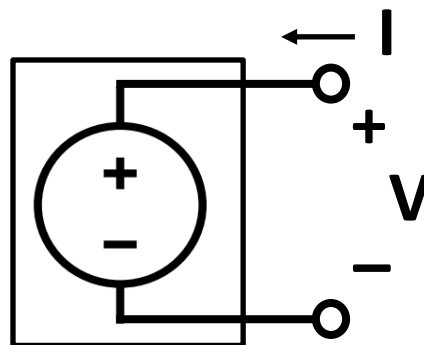
Review of Lesson 1 (e)

- i-v* characteristics / idealized mathematical models of some elements:

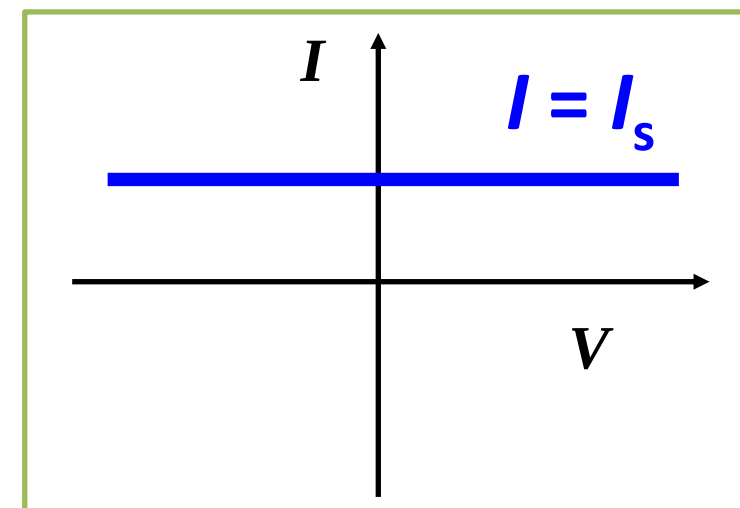
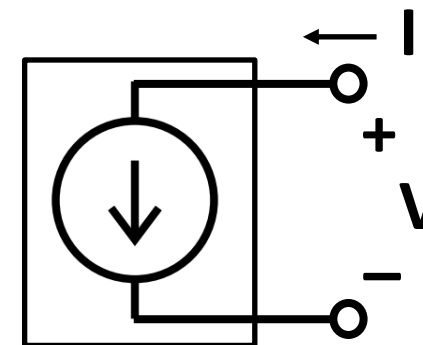
Resistors



Constant V source

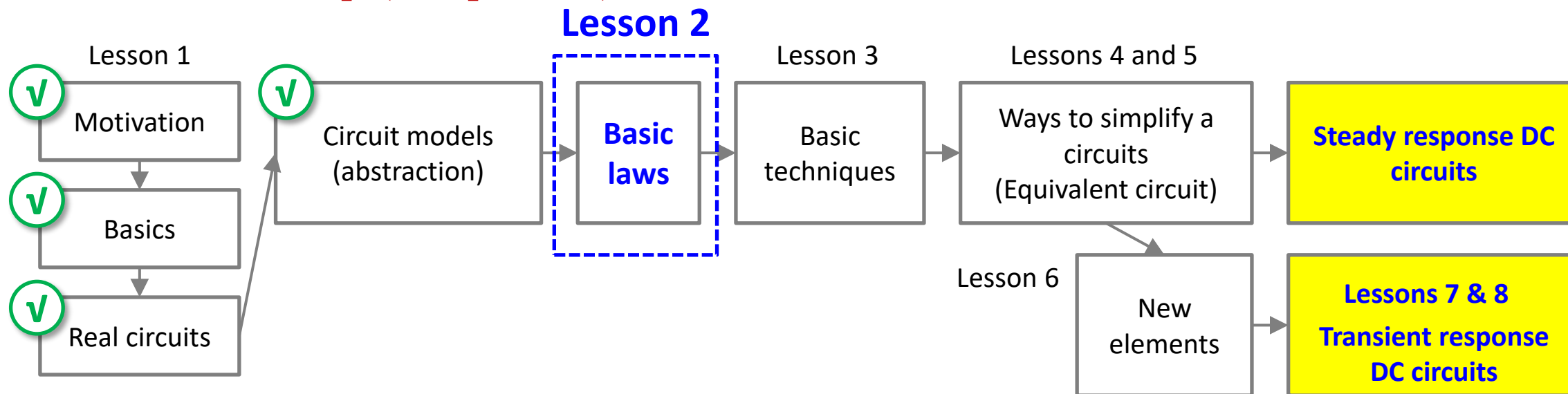


Constant I source



Review of Lesson 1 (f)

- EE104 roadmap (DC portion):



a) The circuits big three

- Ohm's law
- KCL
- KVL

b) Equivalent circuits

- R in series (voltage division)
- R in parallel (current division)
- Y- Δ networks

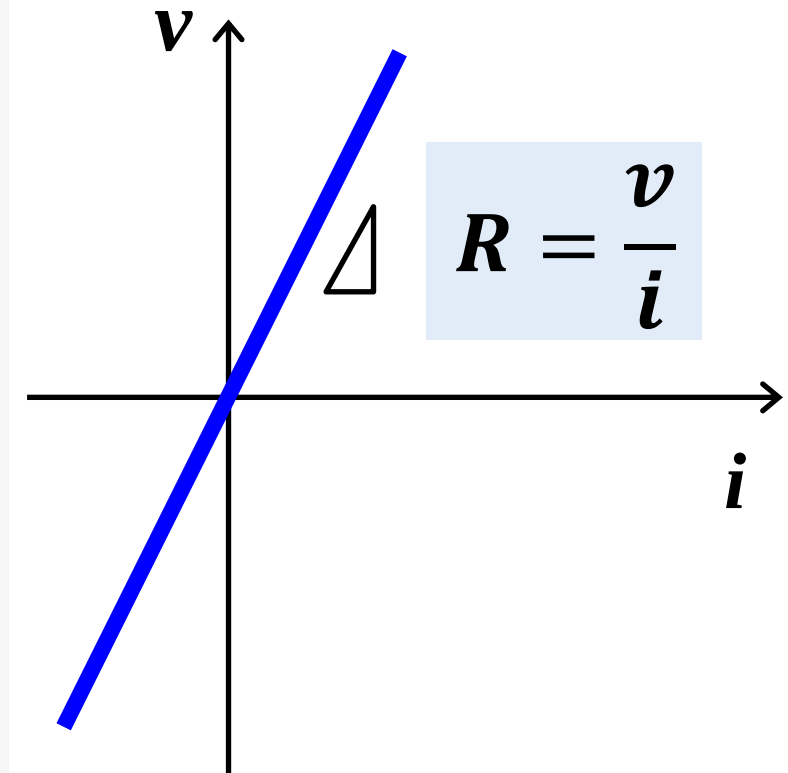
Chapter 2 Basic Laws

2.1-2.2	Ohm's law	Slides 09-20
2.3-2.6	Kirchhoff's Current & voltage Laws (KCL & KVL)	Slides 22-39
2.7	Equivalent circuits	Slides 41-47
2.8-2.10	Series and parallel resistors	Slides 49-62
2.11	Δ -Y and Y- Δ transformations	Slides 64-76
2.12	Summary	Slides 78-79
2.13	Assignments	Slide 80

2.1 Ohm's Law (a)

- Resistance:

- a) Ohm's law states that the voltage v across a conductor is directly proportional to the current i flowing through it, provided all physical conditions and temperature remain constant.
- b) “Directly proportional” means a straight-line relationship in i - v plots.
- c) Resistance (电阻, R): the resistance (R) models the current-resisting behavior of materials and devices, in units of Ohms, Ω , (V/A).
- d) Ohm's Law is the **FOUNDATION** of electronics and electricity.



2.1 Ohm's Law (b)

- Ohm's Law Magic Triangle:

$$i = \frac{v}{R}$$



To calculate i : cover i in the triangle and then you have v over R .

$$R = \frac{v}{i}$$



To calculate R : cover R in the triangle and then you have v over i

$$v = i \cdot R$$



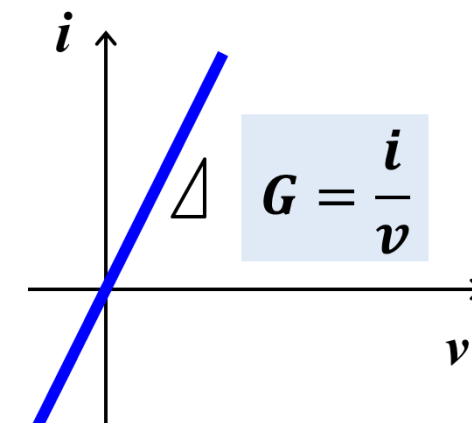
To calculate v : cover v in the triangle and then you have i with R

2.1 Ohm's Law (c)

- Conductance:

- The reciprocal of resistance is conductance (电导, G), which is a measure of how well an element can conduct electric current.
- 1971, the units of conductance has been established as seimans (S).
- Ohm's Law can also be written as follows:

$$G = \frac{1}{R} = \frac{i}{v}$$



$$i = \frac{v}{R} = v \cdot G$$

$$R = \frac{v}{i} = \frac{1}{G}$$

$$v = i \cdot R = \frac{i}{G}$$

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$



$$p = vi = i^2 R = i^2 / G = v^2 G = v^2 / R$$

2.1 Ohm's Law (d)

- Example 01

Example 01: In the circuit shown below, calculate the current i , the conductance G , and the power p .

Solution:

Since the resistor and the voltage source are in parallel, the current is:

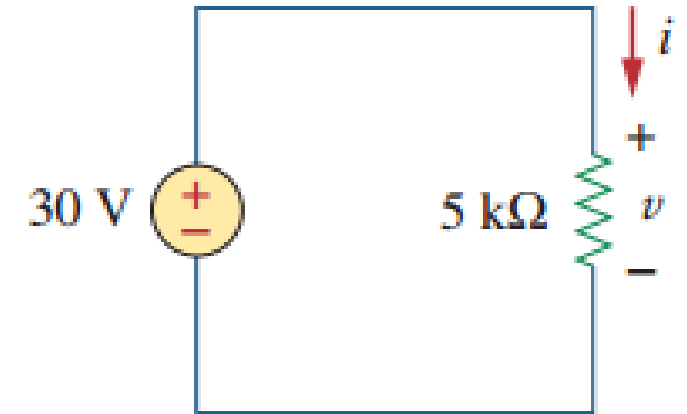
$$i = v/R = 30 / (5 \times 10^3) = 6 \text{ mA}$$

The conductance is:

$$G = 1/R = 1 / (5 \times 10^3) = 0.2 \text{ mS}$$

The power is: $p = vi = 30 \text{ V} \times 6 \text{ mA} = 180 \text{ mW}$

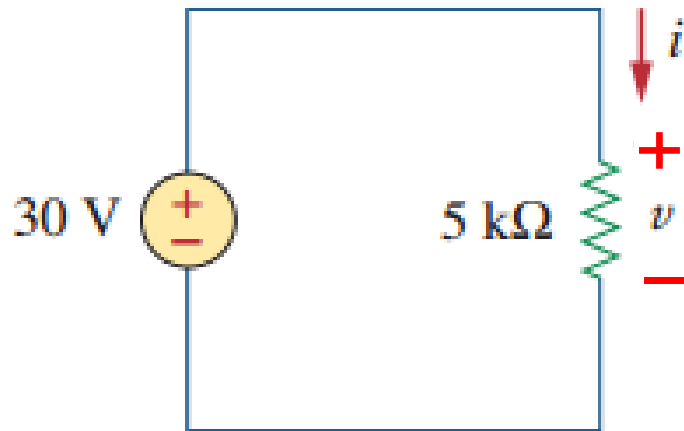
Or: $p = v^2G = (30 \text{ V})^2 \times 0.2 \text{ mS} = 180 \text{ mW}$ Or: $p = i^2R = (6 \times 10^{-3} \text{ A})^2 \times 5 \times 10^3 \Omega = 0.18 \text{ W}$



2.1 Ohm's Law (e)

- Ohm's law & reference directions

Associated reference directions for R

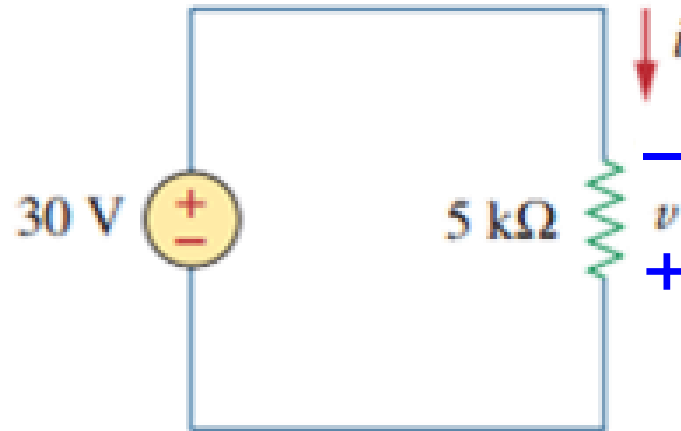


$$v = 30 \text{ V},$$

$$i = 6 \text{ mA}, R = 5 \text{ k}\Omega$$

$$v = iR$$

Nonassociated reference directions for R



$$v = -30 \text{ V},$$

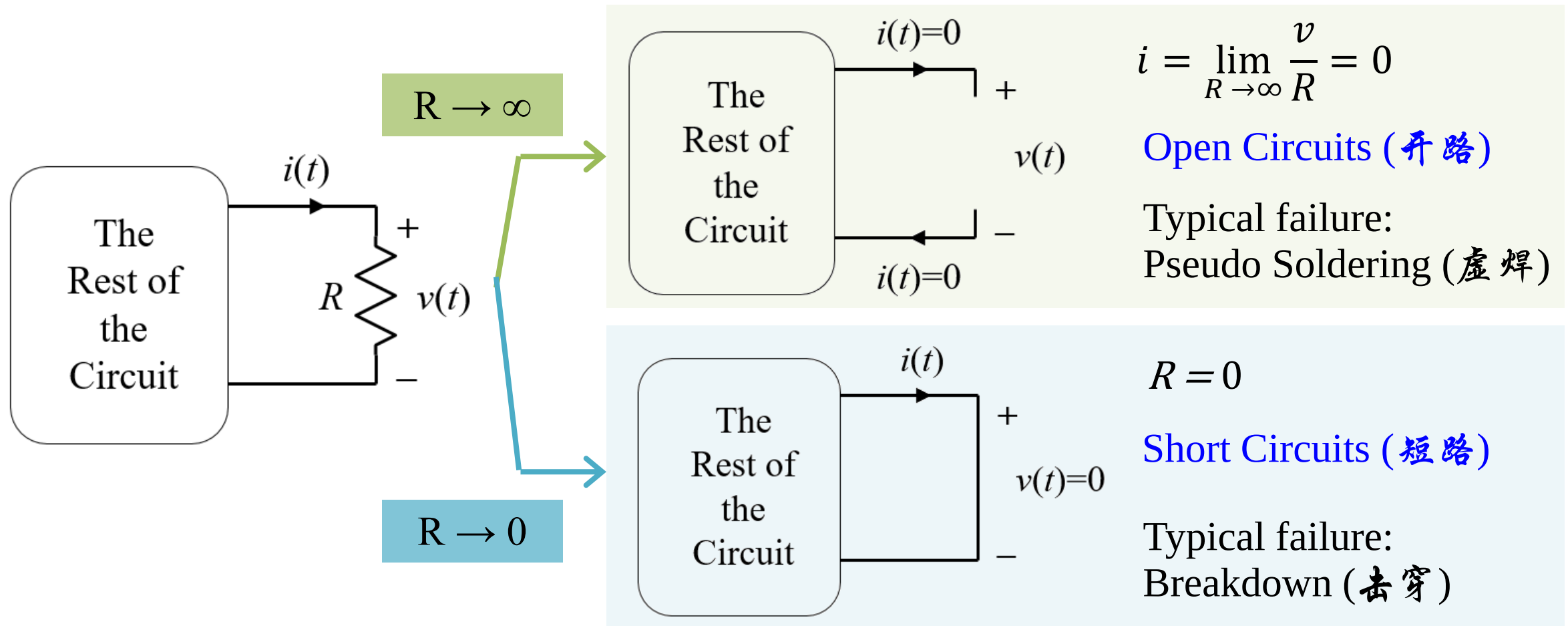
$$i = 6 \text{ mA}, R = 5 \text{ k}\Omega$$

$$v = -iR$$

- Let's revisit Example 01.
- Depending on the reference directions, the expression of Ohm's law can be different.
- Once you find a convenient way of selecting reference directions, be consistent.
- It is suggested to use the passive sign convention as much as possible.

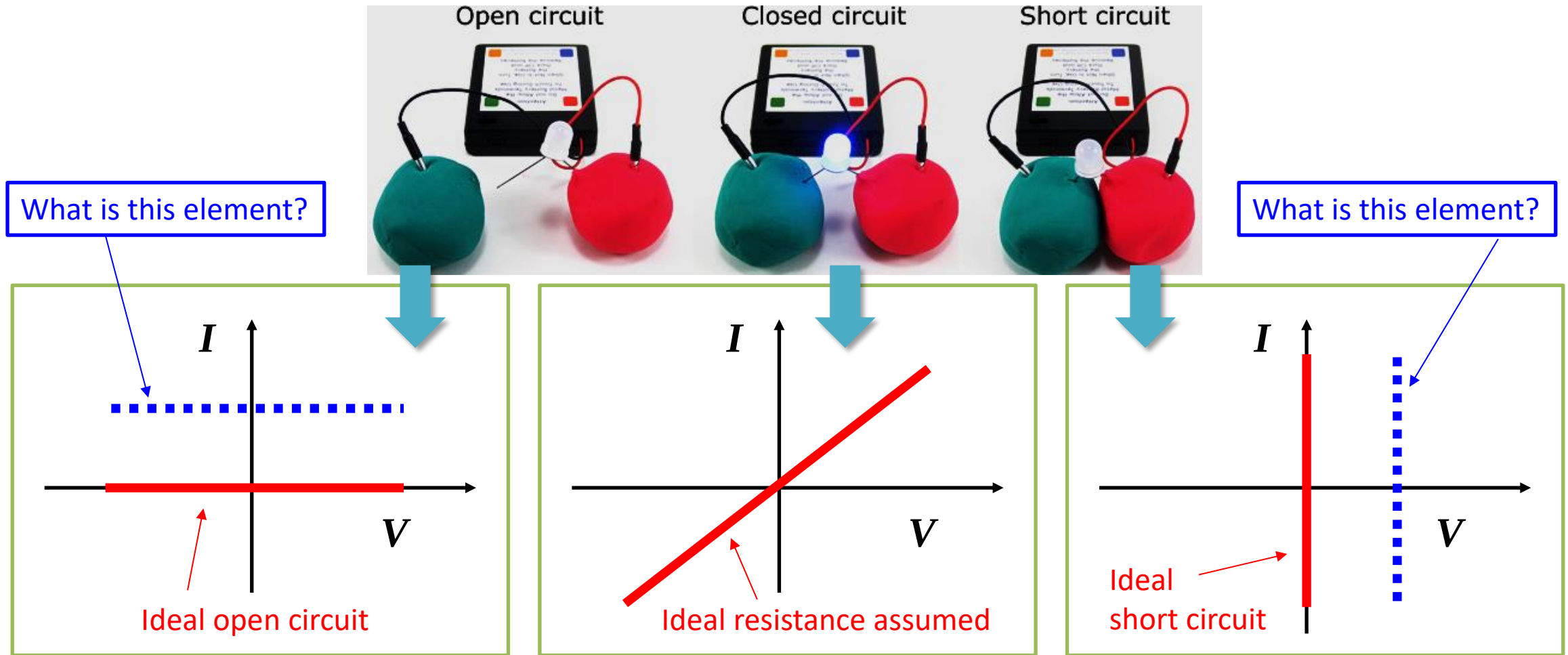
2.1 Ohm's Law (f)

- Open and short circuits



2.1 Ohm's Law (g)

- i - v characteristics / mathematic models of open and short circuits.

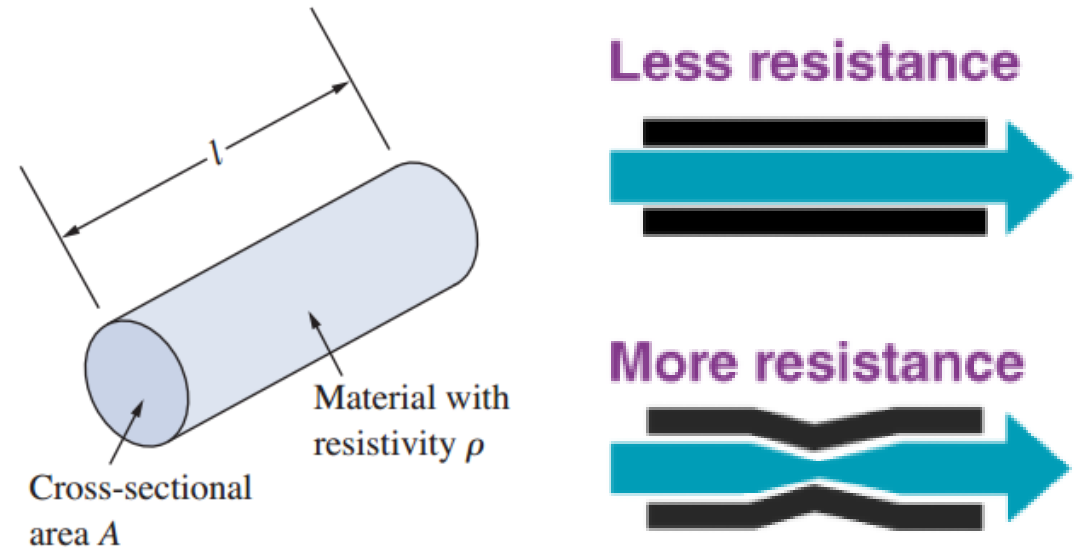


2.2 Resistance and Resistivity (a)

- Resistivity:

- The resistance of a resistor is determined by its dimension as well as the resistivity of the material composing the resistor, provided all physical conditions and temperature remain constant.
- The **electrical resistivity** (电阻率, ρ) is a measure of how strongly a material opposes the flow of electric current, which is also be known as its specific electrical resistance of a material.
- The SI unit of ρ is $\Omega \cdot \text{m}$

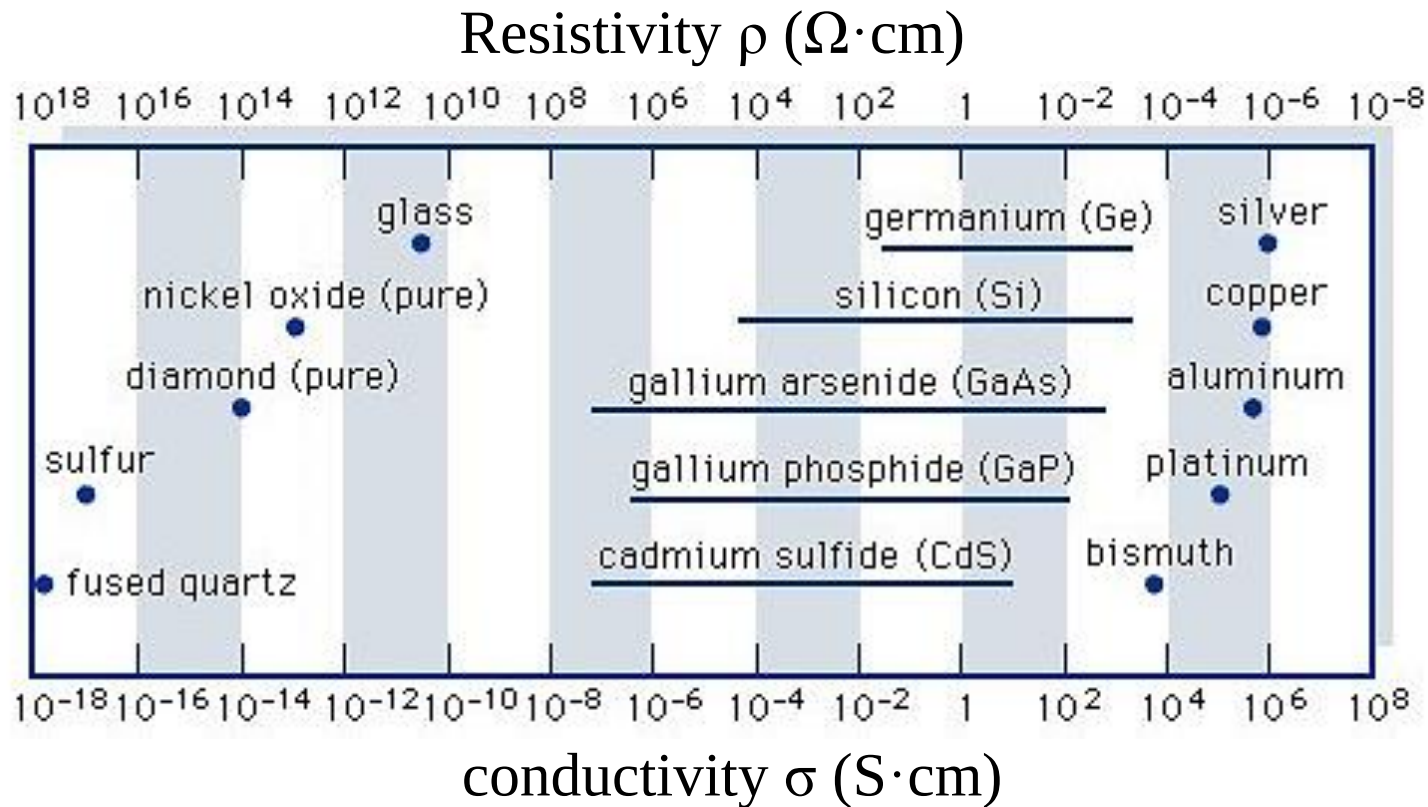
$$R = \rho \frac{\ell}{A}$$



Similar to water flows.

2.2 Resistance and Resistivity (b)

- Resistivity of common materials:

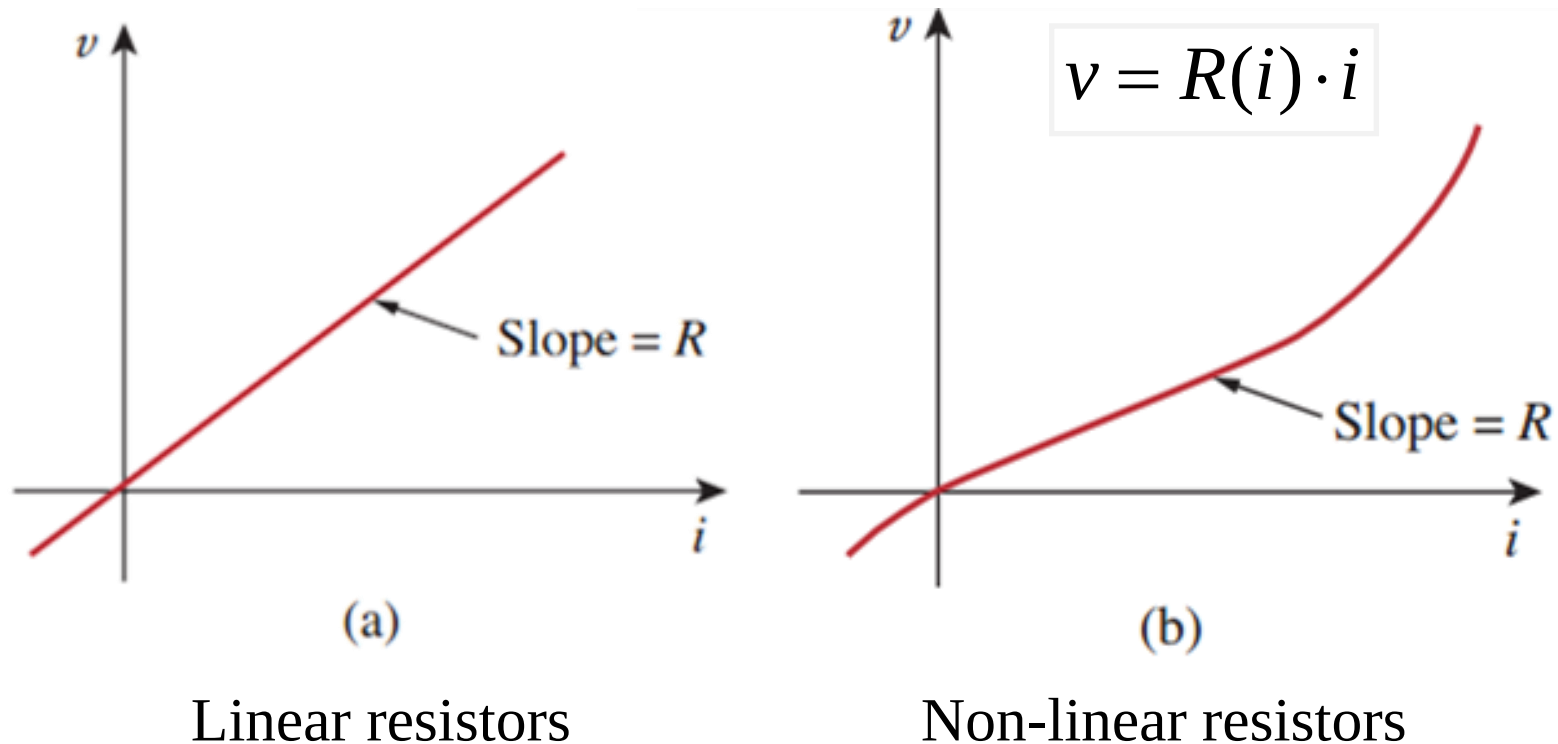


Similar to $G = 1/R$, $\sigma = 1/\rho$

- Metals are often **conductors** thanks to their low ρ .
- Insulators** exhibit high ρ so they are considered as insulating.
- The ρ of typical **semiconductors** lies between those of conductors and those of insulators (or featuring a tunable electrical conductivity).
- Many traditional insulators are now semiconductors, including Diamond, Nickel oxide, AlN, etc. Why?

2.2 Resistance and Resistivity (c)

- Nonlinearity of resistors:



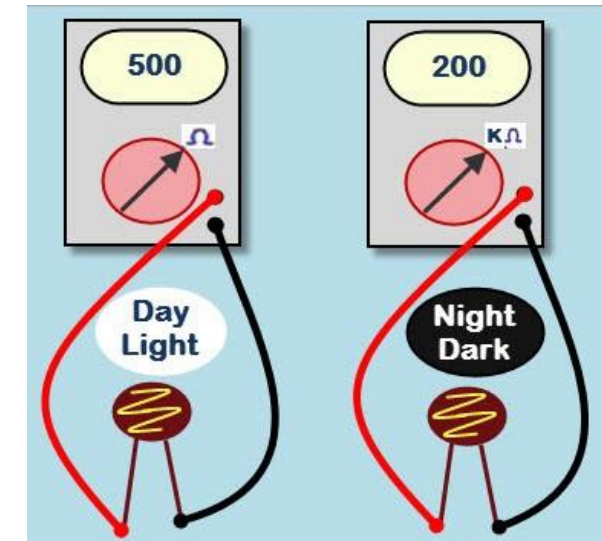
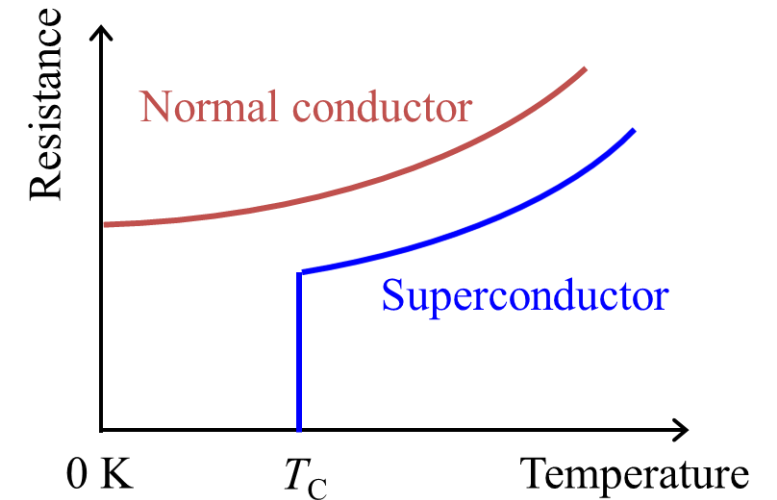
To think about:

Is the resistance of the filament of a bulb constant?

How does it change with the temperature?

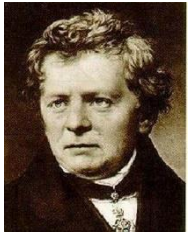
2.2 Resistance and Resistivity (d)

- Some other factors can affect the ρ :
- Ohm's law states that the voltage v across a conductor is directly proportional to the current i flowing through it, **provided all physical conditions and temperature remain constant.**
 - The resistance of a resistor is determined by its dimension as well as the resistivity of the material composing the resistor, **provided all physical conditions and temperature remain constant.**
 - The resistivity of a material can be affected by other factors, including temperature, light, etc.



Interesting To Know

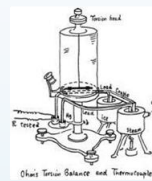
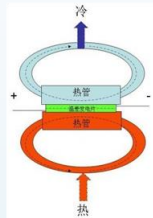
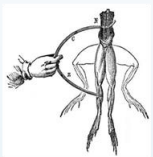
Georg Simon Ohm



1787.03.16 – 1854.07.06

German Physicist and mathematician
Ohm's Law

Sources

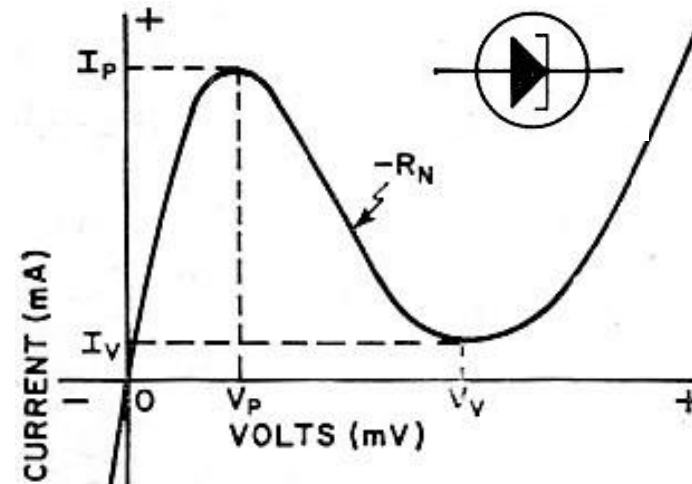


Fourier law of heat conduction: the heat flux vector is proportional to the negative vector gradient of temperature.

***Henry Cavendish:** qualitatively described the relationship between voltage, resistance and current in an electrical circuit

Ohm's Law

Negative resistance



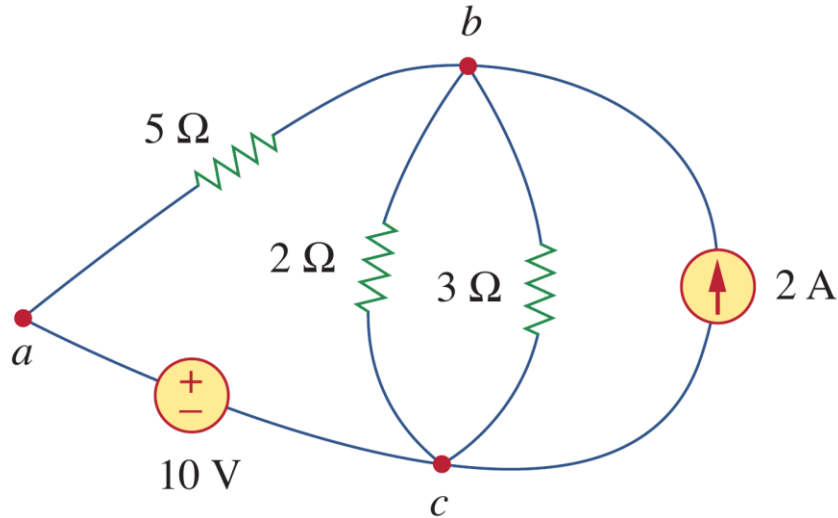
Tunneling diodes:

Invented by Leo Esaki in 1957, who received the Nobel prize in Physics in 1973 for discovering the electron tunneling effect in these diodes

Chapter 2 Basic Laws

2.1-2.2	Ohm's law	Slides 09-20
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2.3 Network topology (a)



5 branches, $b = 5$

3 nodes, $n = 3$

3 meshes, $m = 3$

6 loops

$$b = m + n - 1$$

Branch
(支路)

A branch represents a single element such as a voltage source or a resistor.

Node
(节点)

A node is the point of connection between two or more branches.

Loop
(回路)

A loop is a closed path in a circuit.

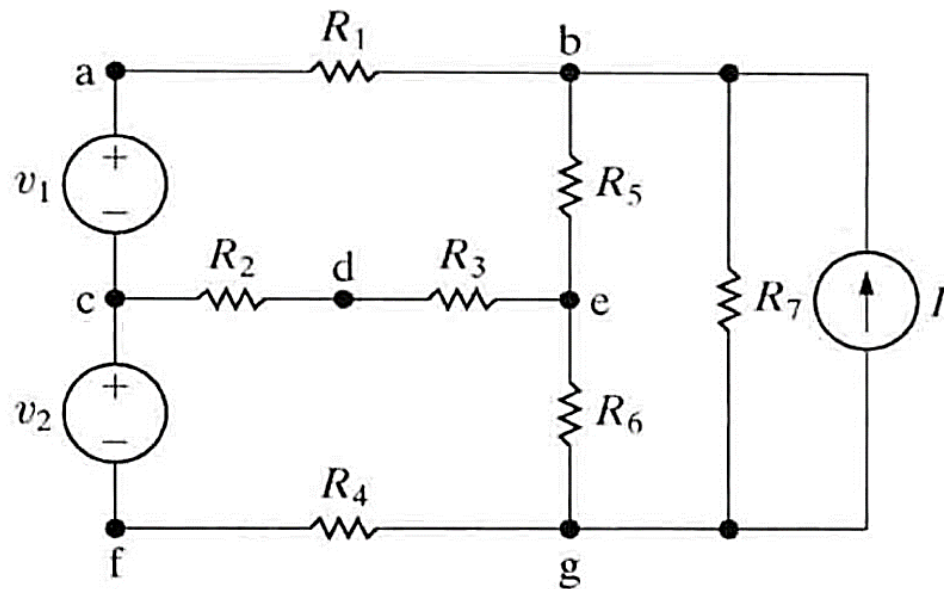
Mesh
(网孔)

A mesh is a loop which does not contain any other loops within it.

2.3 Network topology (b)

• Example 02

Example 02: Find the numbers of (a) nodes, (b) branches, (c) meshes, and identify (d) 2 loops that are not meshes.



$$b = m + n - 1$$

Solution:

(a)	Nodes	$n = 7$	a, b, c, d, e, f, g
(b)	Branches	$b = 10$	$v_1, v_2, R_1, R_2, R_3, R_4, R_5, R_6, R_7, I$
(c)	Meshes	$m = 4$	(1) $v_1 - R_1 - R_5 - R_3 - R_2$ (2) $v_2 - R_2 - R_3 - R_6 - R_4$ (3) $R_5 - R_7 - R_6$ (4) $R_7 - I$
(d)	2 loops (not meshes)	--	(1) $v_1 - R_1 - R_5 - R_6 - R_4 - v_2$ (2) $I - R_5 - R_6$

2.4 Kirchhoff's Current Law KCL (a)

- KCL:

- a) KCL (基尔霍夫电流定律): The algebraic sum of currents entering a node (or a closed boundary-闭合界面) is zero. In other words, the sum of the currents entering a node is equal to that leaving the node.
- b) Principle of the conservation of charge: The algebraic sum of charges within a system cannot change.

$$0 = \frac{dq}{dt} = \sum_{n=1}^N i_n$$

OR

$$\sum i_{\text{in}} = \sum i_{\text{out}}$$

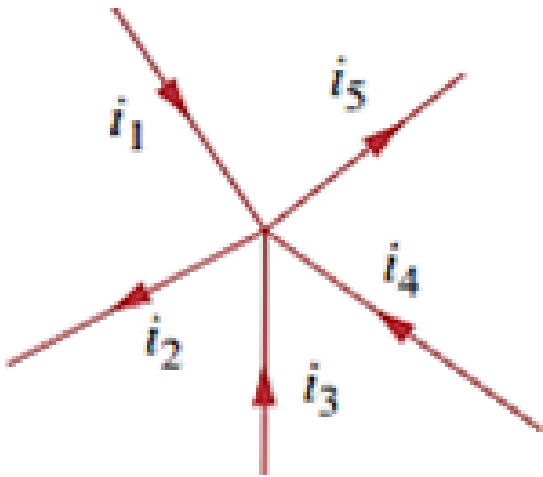


Fig. a

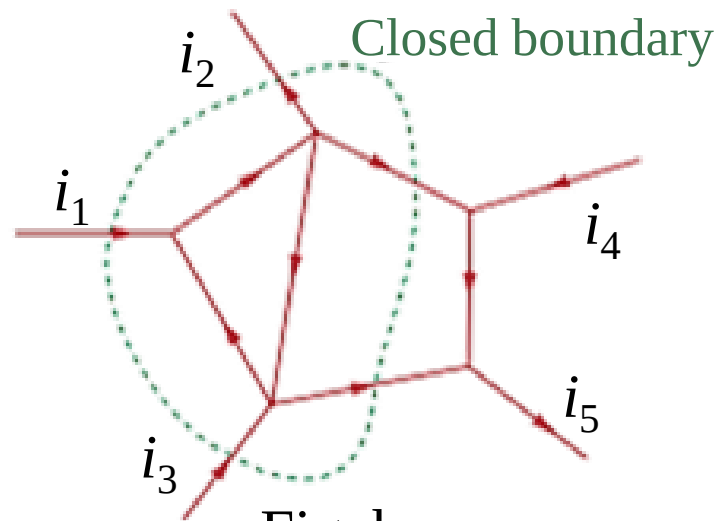


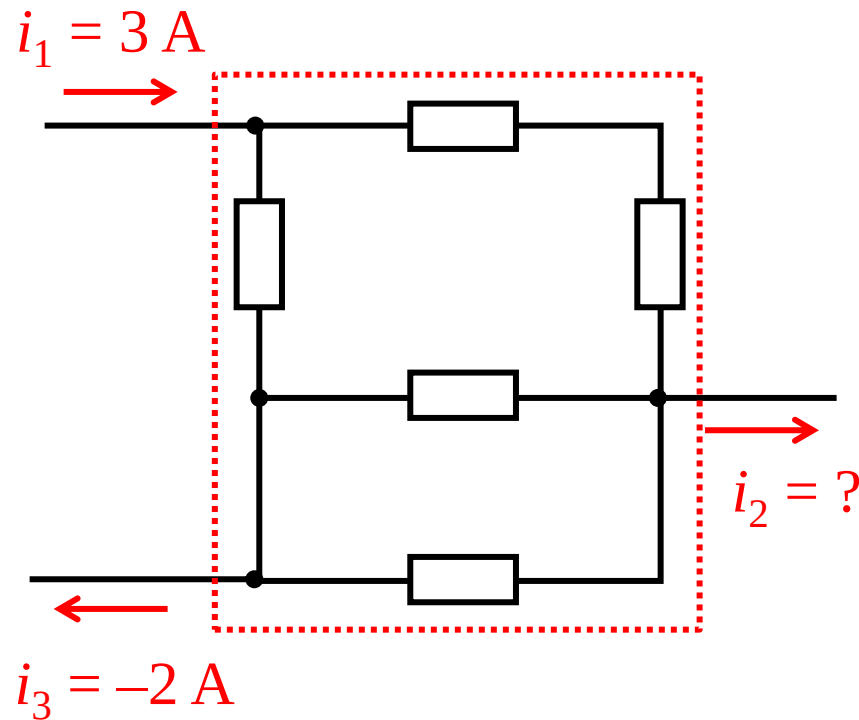
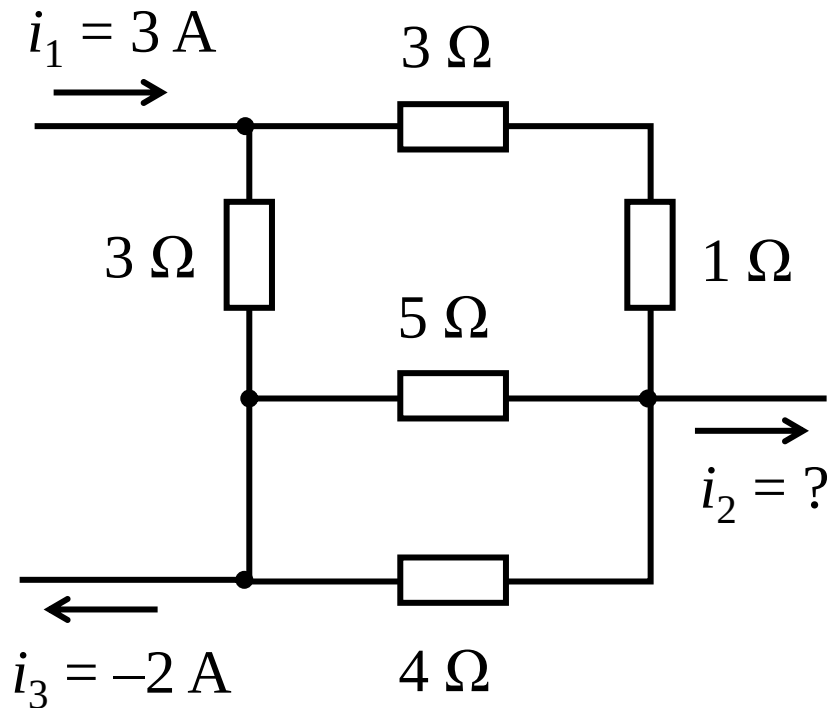
Fig. b

$i_1 - i_2 + i_3 + i_4 - i_5 = 0$ in Figs. a and b
+ when the reference current direction is into the node or the closed boundary.
– when the reference current direction is out from the node or the closed boundary.

2.4 Kirchhoff's Current Law KCL (b)

- Example 03:

Example 03a: Calculate i_2 in the circuit shown below.



Solution:

$$i_1 - i_2 - i_3 = 0$$

$$3 - i_2 - (-2) = 0$$

$$i_2 = 5 \text{ A (out)}$$

$$\text{Or } -i_1 + i_2 + i_3 = 0$$

$$-3 - i_2 + (-2) = 0$$

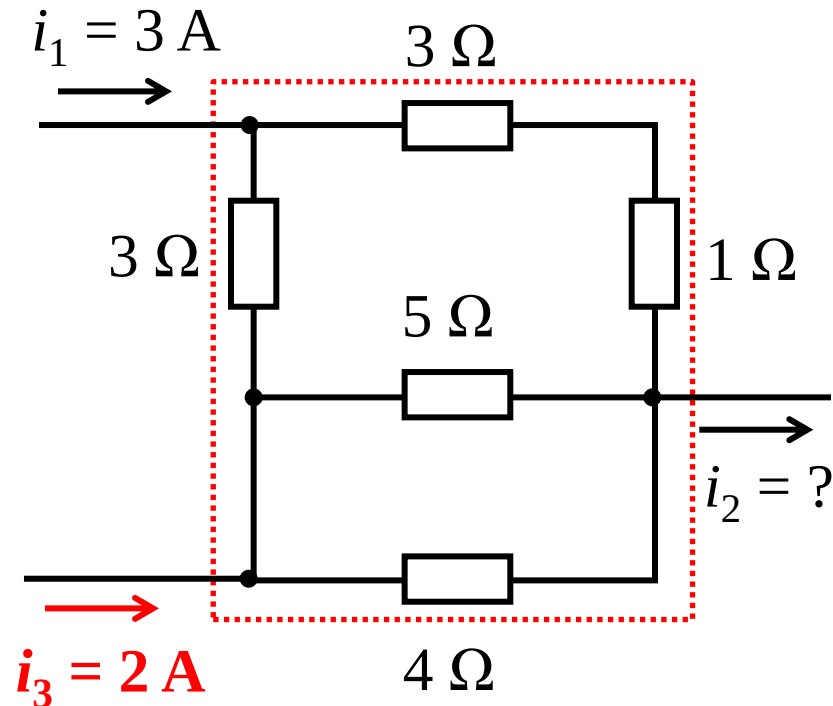
$$i_2 = 5 \text{ A (out)}$$

2.4 Kirchhoff's Current Law KCL (c)

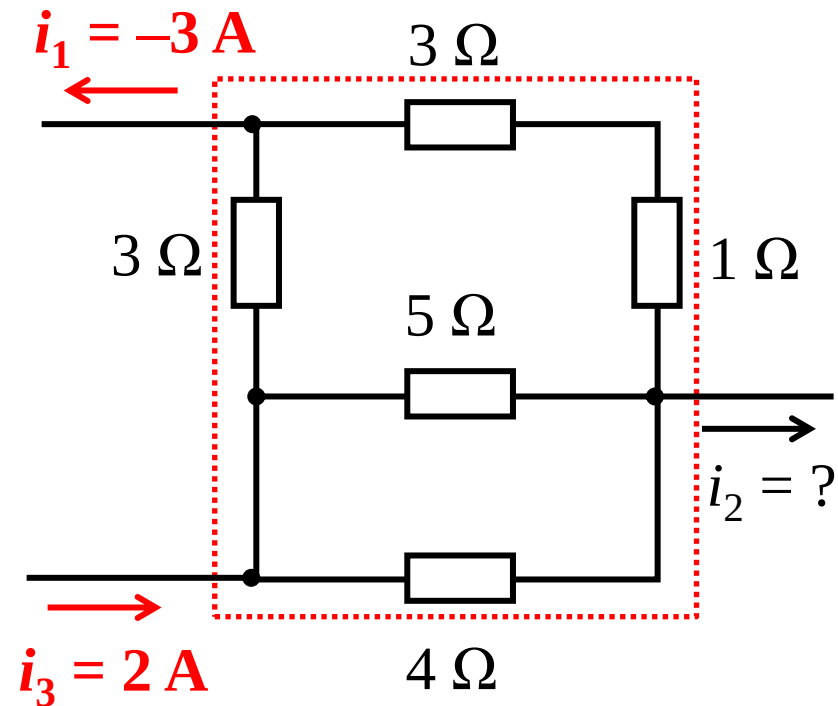
- Example 03:

Examples 03b and c: Calculate i_2 in the circuit shown below.

$$i_1 - i_2 + i_3 = 0 \rightarrow 3 - i_2 + 2 = 0 \rightarrow i_2 = 5 \text{ A out}$$



$$-i_1 - i_2 + i_3 = 0 \rightarrow -(-3) - i_2 + 2 = 0 \rightarrow i_2 = 5 \text{ A out}$$

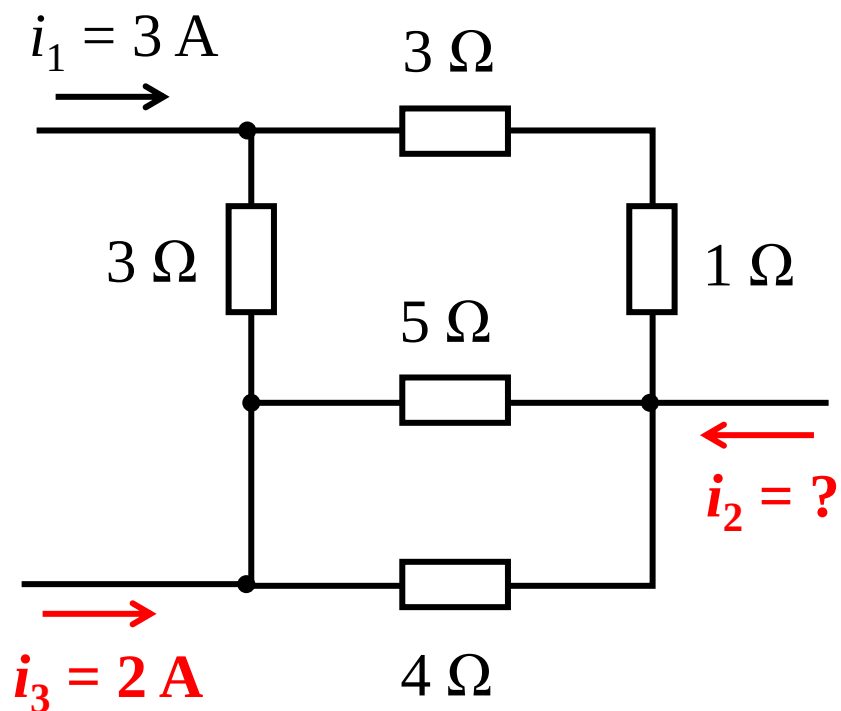


2.4 Kirchhoff's Current Law KCL (d)

- Example 03:

Examples 03d: Calculate i_2 in the circuit shown below.

$$i_1 + i_2 + i_3 = 0 \rightarrow 3 + i_2 + 2 = 0 \rightarrow i_2 = -5 \text{ A out}$$



Q	i_1	i_3	i_2	Results (i_2)
03a	→	←	→	$i_2 = 5 \text{ A, out}$
03b	→	→	→	$i_2 = 5 \text{ A, out}$
03c	←	→	→	$i_2 = 5 \text{ A, out}$
03d	→	→	←	$i_2 = -5 \text{ A, out}$

What can you learn from these?

2.4 Kirchhoff's Current Law KCL (e)

- Tips when using KCL:

- a) Write your KCL equation based reference current directions, not the real ones.
- b) When you write your KCL equation, the currents flowing into the node or the closed boundary can be considered as positive, and the current flowing out can be considered as negative in the equation ($i_1 + i_2 + i_3 + \dots + i_N = 0$). Of course you can adopt the other way if you really wish so.
- c) KCL represents the conservation of charge, and is true in lumped circuits or under lumped circuit abstraction (LCA).

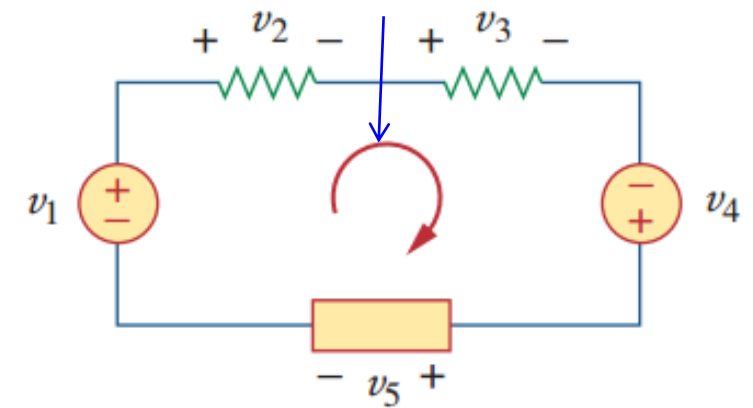
2.5 Kirchhoff's voltage Law KVL (a)

- KVL:

$$\sum_{m=1}^M w_m = 0 = \sum_{m=1}^M qv_m \Rightarrow \boxed{\sum_{m=1}^M v_m = 0}$$

- a) KVL (基尔霍夫电压定律) states that: the algebraic sum of all voltages around a loop is zero, or Sum of voltage drops = Sum of voltage rises, which is based on the conservation of energy.
- b) Consider to move a charge along the loop in the right figure, when the charge is moved back to the starting point, the net work experienced (applied) must equal to zero; otherwise the potential at the starting point would have multiple values which should NOT occur.

Loop direction (绕行方向)



$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

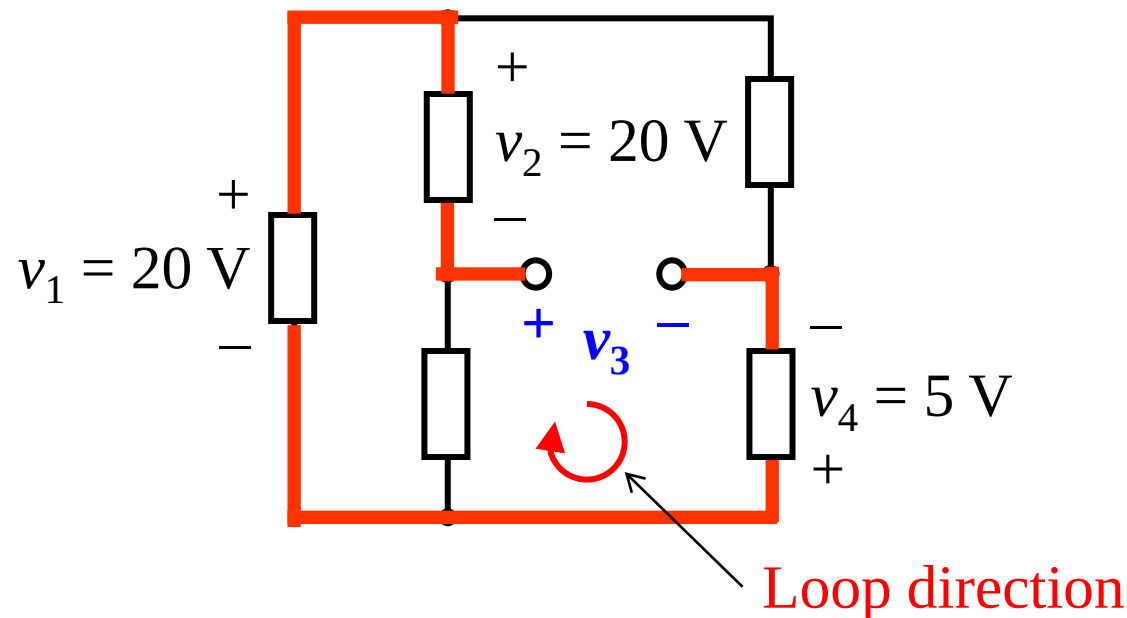
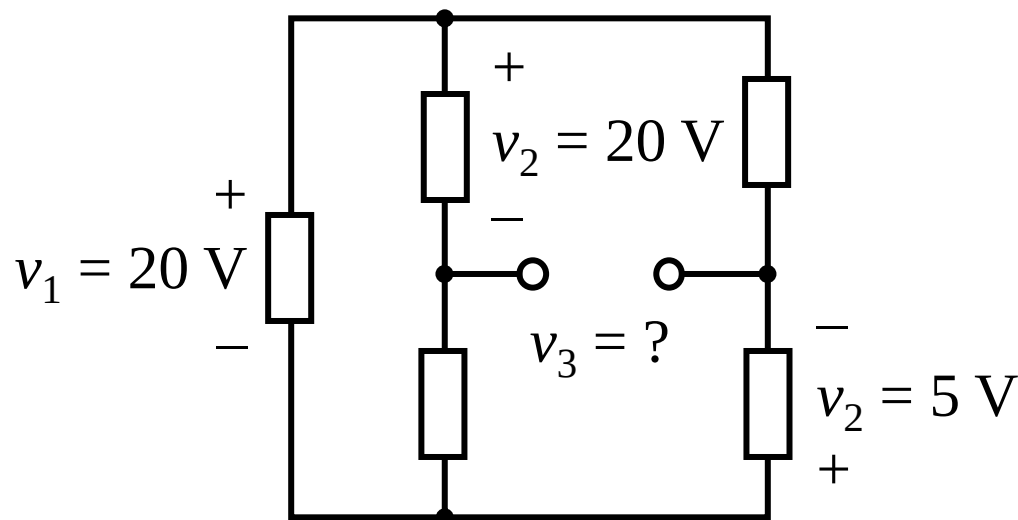
+ when the potential decreases across the device along the loop direction. (associated)

– when the potential increases across the device along the loop direction. (nonassociated)

2.5 Kirchhoff's voltage Law KVL (b)

- Example 04 a to c:

Example 04a: Calculate v_3 in the circuit shown below.

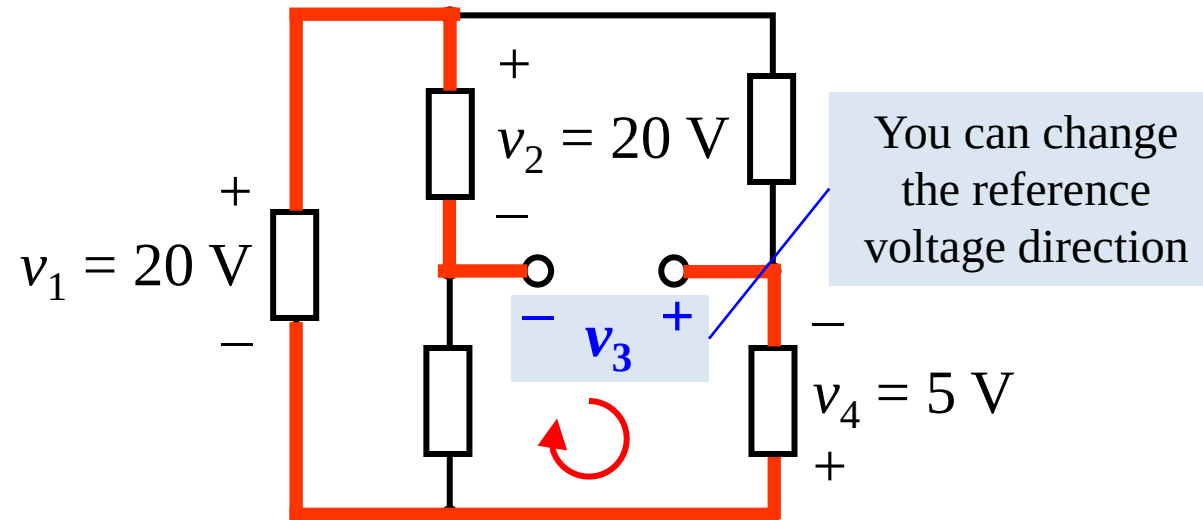
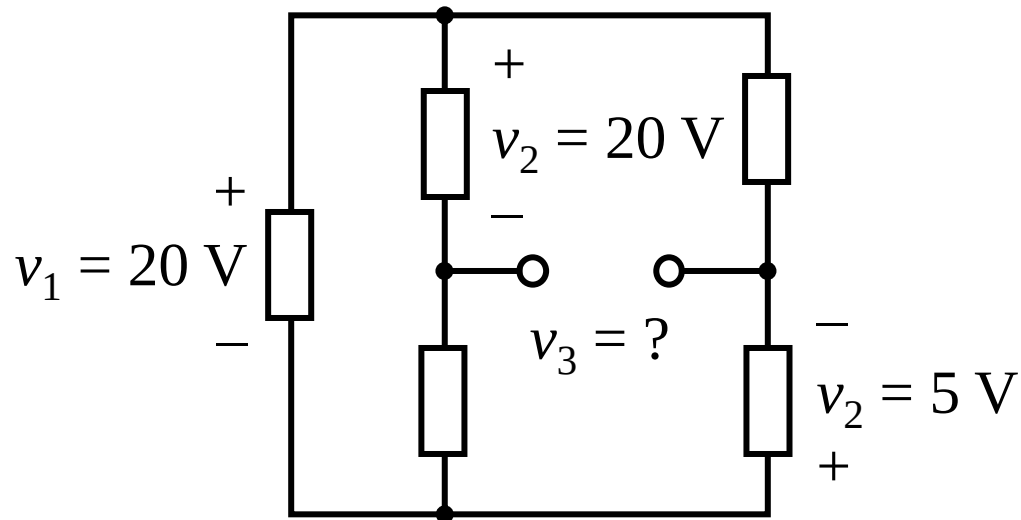


Solution I: $-v_1 + v_2 + v_3 - v_4 = 0 \rightarrow v_3 = v_1 + v_4 - v_2 \rightarrow v_3 = 5 \text{ V}$

2.5 Kirchhoff's voltage Law KVL (c)

- Example 04 a to c:

Example 04b: Calculate v_3 in the circuit shown below.



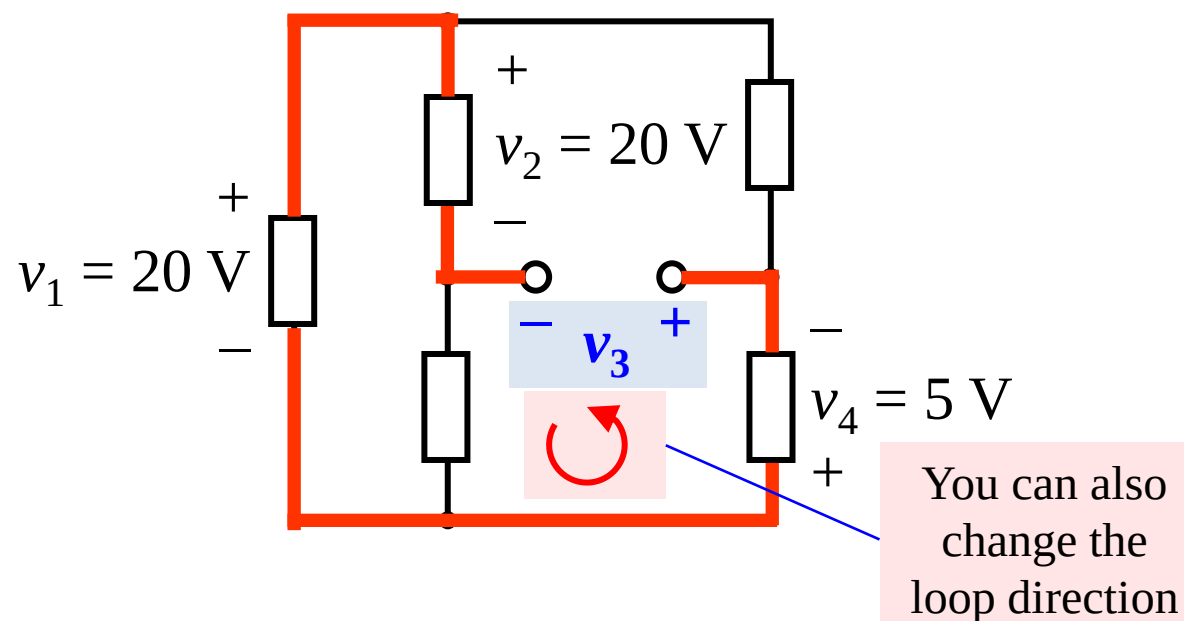
Solution II: $-v_1 + v_2 - v_3 - v_4 = 0 \rightarrow v_3 = v_2 - v_1 - v_4 \rightarrow v_3 = -5 \text{ V}$

2.5 Kirchhoff's voltage Law KVL (d)

- Example 04 a to c:

Example 04c: Calculate v_3 in the circuit shown below.

Q	V_3 Ref	Loop direction	Results (v_3)
04a	+ -	CW (顺)	$v_3 = 5 \text{ V}$, +-
04b	- +	CW (顺)	$v_3 = -5 \text{ V}$, +-
04c	- +	ACW (逆)	$v_3 = -5 \text{ V}$, +-



Solution III: $+v_1 - v_2 + v_3 + v_4 = 0 \rightarrow v_3 = v_2 - v_1 - v_4 \rightarrow v_3 = -5 \text{ V}$

2.5 Kirchhoff's voltage Law KVL (e)

- Tips when using KVL:

- Write your KVL equation based on reference voltage directions, not real ones.
- The selection of loop direction is up to you, which can be either clockwise (顺时针) or counterclockwise (逆时针), as the only thing that matters is your consistency.
- You can use “potential increases or decreases” to decide the terms to be plus or minus in the KVL. Or you can use “being associated or nonassociated” to determine: voltages associated to the loop direction can be positive, and voltages nonassociated to the loop direction can be negative in the KVL equation ($v_1 + v_2 + v_3 + \dots + v_N = 0$).
- KVL represents the conservation of charge, and is true in lumped circuits (or under lumped circuit abstraction (LCA)).

2.6 KCL + KVL (a)

- KCL and KVL are of high importance:

a) Ohm's law, KCL and KVL are “the circuits big three”!

b) With Kirchhoff's two laws, we have a sufficient, powerful set of tools for analyzing a large variety of electric circuits.

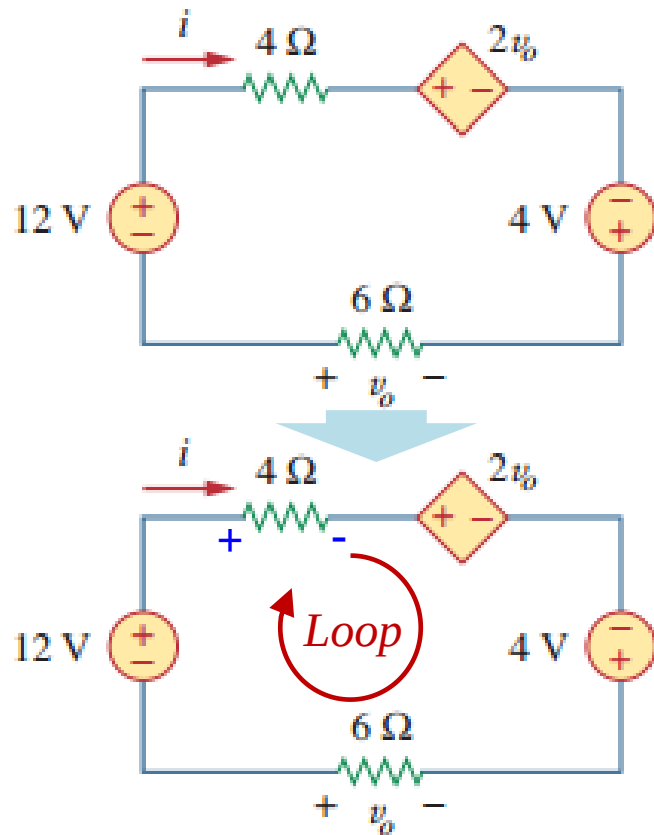
c) KCL is also called The Junction Rule

d) KVL is also called The Loop Rule

2.6 KCL + KVL (b)

- Example 05:

Example 05a: Determine v_0 and i in the circuit shown below.



Solution I (clockwise loop direction):

$$\begin{cases} \text{KVL: } -12 + 4i + 2v_0 - 4 - v_0 = 0 & (1) \\ \text{Ohm's Law: } v_0 = -6i & (2) \end{cases}$$

Based on (1) and (2):

$$-12 + 4i - 12i - 4 + 6i = 0$$

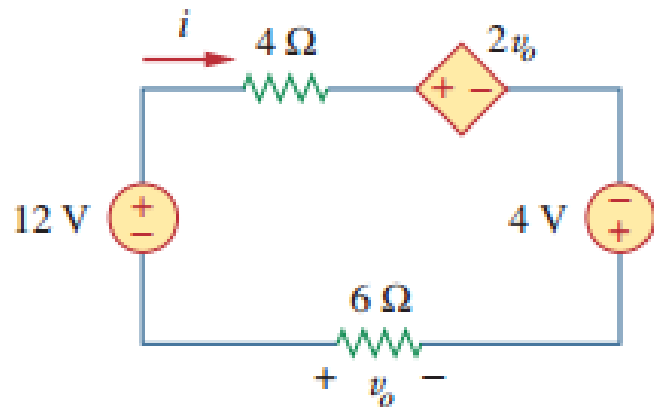
$$-2i = 16$$

$$i = -8 \text{ A}, \quad v_0 = 48 \text{ V}$$

2.6 KCL + KVL (c)

- Example 05:

Example 05b: Determine v_0 and i in the circuit shown below.



Solution II (anticlockwise loop direction):

$$\begin{cases} \text{KVL: } +12 - 4i - 2v_0 + 4 + v_0 = 0 & (1) \\ \text{Ohm's Law: } v_0 = -6i & (2) \end{cases}$$

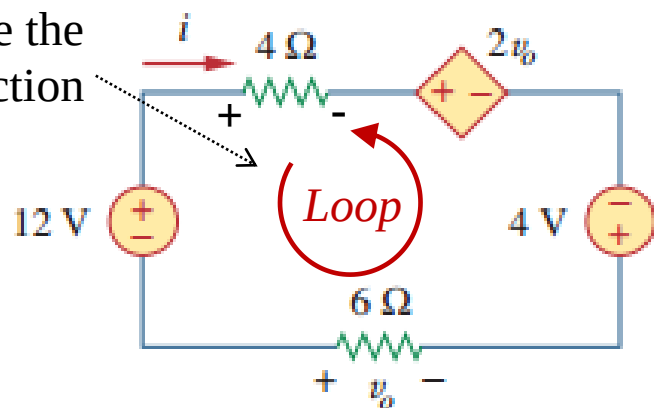
Based on (1) and (2):

$$12 - 4i + 12i + 4 - 6i = 0$$

$$2i = -16$$

$$i = -8 \text{ A}, \quad v_0 = 48 \text{ V}$$

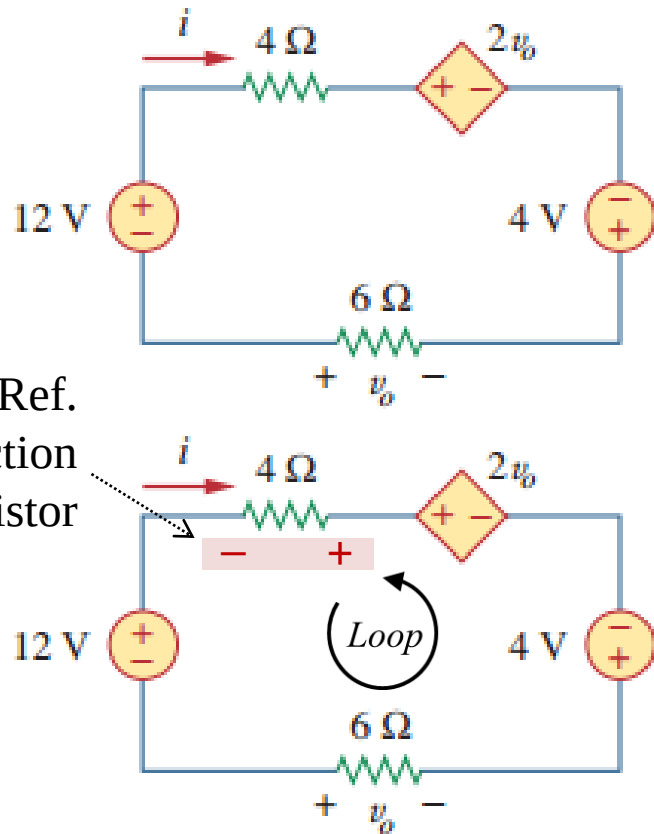
Change the
loop direction



2.6 KCL + KVL (d)

- Example 05:

Example 05c: Determine v_0 and i in the circuit shown below.



Solution III:

$$\begin{cases} \text{KVL: } +12 + (-4i) - 2v_0 + 4 + v_0 = 0 & (1) \\ \text{Ohm's Law: } v_0 = -6i & (2) \end{cases}$$

Based on (1) and (2):

$$12 - 4i - 12i + 4 - 6i = 0$$

$$2i = -16$$

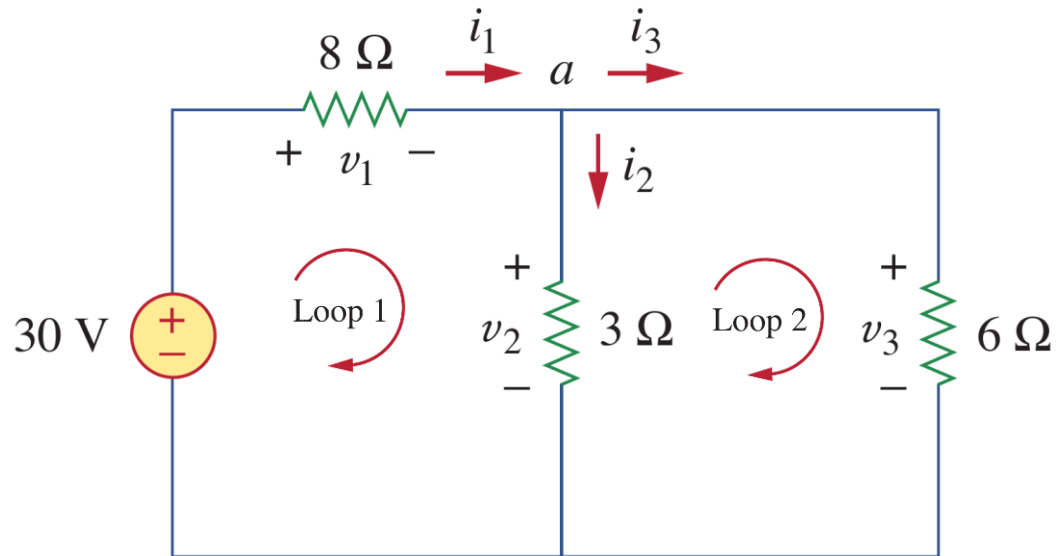
$$i = -8 \text{ A}, \quad v_0 = 48 \text{ V}$$

2.6 KCL + KVL (e)

- Example 06:

Example 06: Find currents and voltages in the following circuit.

Solution I (clockwise loop direction):



$$\text{KVL} \begin{cases} \text{Loop 1: } -30 + v_1 + v_2 = 0 \\ \text{Loop 2: } -v_2 + v_3 = 0 \end{cases}$$

$$\text{KCL: } i_1 - i_2 - i_3 = 0$$

$$\text{OHM: } v_1 = 8i_1, \quad v_2 = 3i_2, \quad v_3 = 6i_3$$

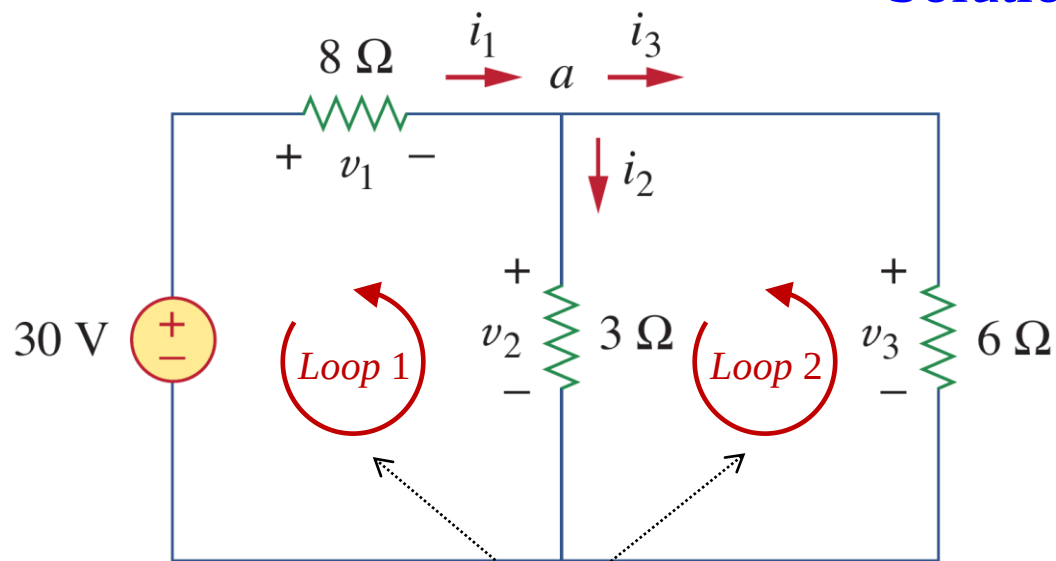
$$\begin{aligned} \text{Then we have: } i_1 &= 3 \text{ A} & i_2 &= 2 \text{ A} & i_3 &= 1 \text{ A} \\ v_1 &= 24 \text{ V} & v_2 &= 6 \text{ V} & v_3 &= 6 \text{ V} \end{aligned}$$

2.6 KCL + KVL (f)

- Example 06:

Example 06: Find currents and voltages in the following circuit.

Solution II (anticlockwise loop direction):



Change the loop direction

$$\text{KVL} \begin{cases} \text{Loop 1: } + 30 - v_1 - v_2 = 0 \\ \text{Loop 2: } + v_2 - v_3 = 0 \end{cases}$$

$$\text{OHM: } v_1 = 8i_1, \quad v_2 = 3i_2, \quad v_3 = 6i_3$$

$$\text{KCL: } i_1 - i_2 - i_3 = 0$$

$$\begin{aligned} \text{Then we have: } i_1 &= 3 \text{ A} & i_2 &= 2 \text{ A} & i_3 &= 1 \text{ A} \\ v_1 &= 24 \text{ V} & v_2 &= 6 \text{ V} & v_3 &= 6 \text{ V} \end{aligned}$$

Chapter 2 Basic Laws

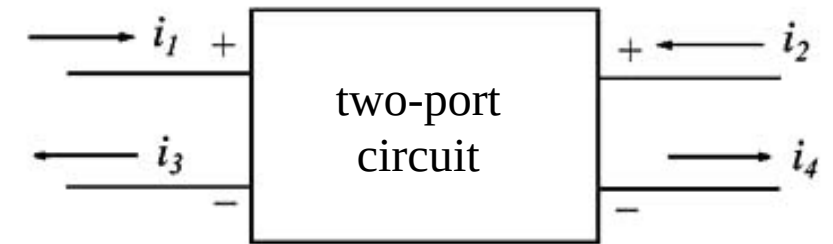
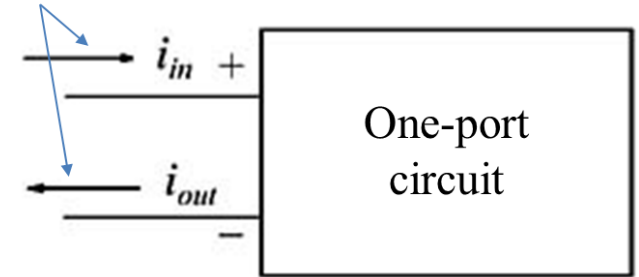
2.1-2.2	Ohm's law	Slides 09-20
2.3-2.6	Kirchhoff's Current & voltage Laws (KCL & KVL)	Slides 22-39
2.7	Equivalent circuits	Slides 41-47
2.8-2.10	Series and parallel resistors	Slides 49-62
2.11	Δ -Y and Y- Δ transformations	Slides 64-76
2.12	Summary	Slides 78-79
2.13	Assignments	Slide 80

2.7 Equivalent circuits (a)

- Equivalent circuits

- a) An **equivalent circuit** is a simplified model of an existing circuit, which can greatly simplify the analysis of the original circuit.
- b) Two circuits (or sources) are said to be **equivalent** (等效的) if they have **the same V-I relationship** (伏安特性) at their ports (端口).
- c) One-port network: generally a network has two pairs of terminals; one pair is denoted as the input terminals, and the other as the output terminals. One-port circuits are usually found at the beginning and the end of a circuit.

Terminals 1 and 2

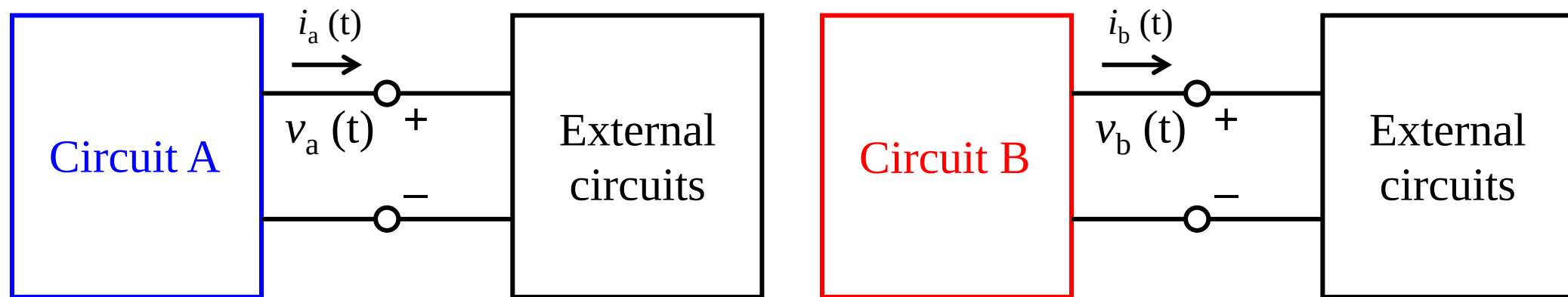


An one-port circuit has two terminals
A two-port circuit has four terminals
A three-port circuit has six terminals

2.7 Equivalent circuits (b)

- Equivalent circuits

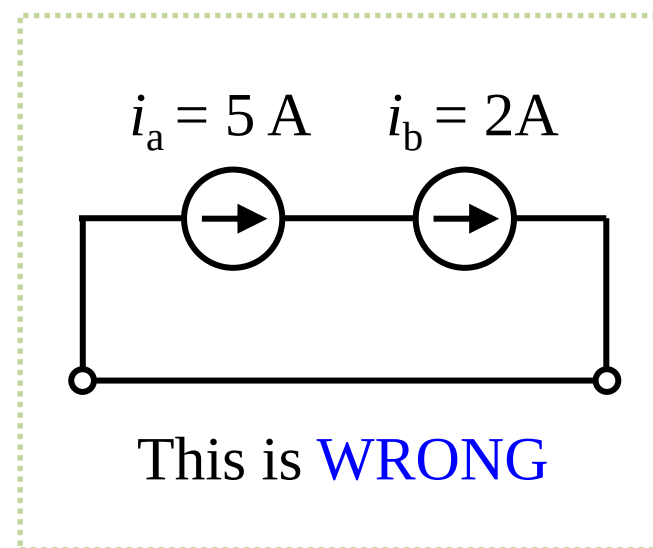
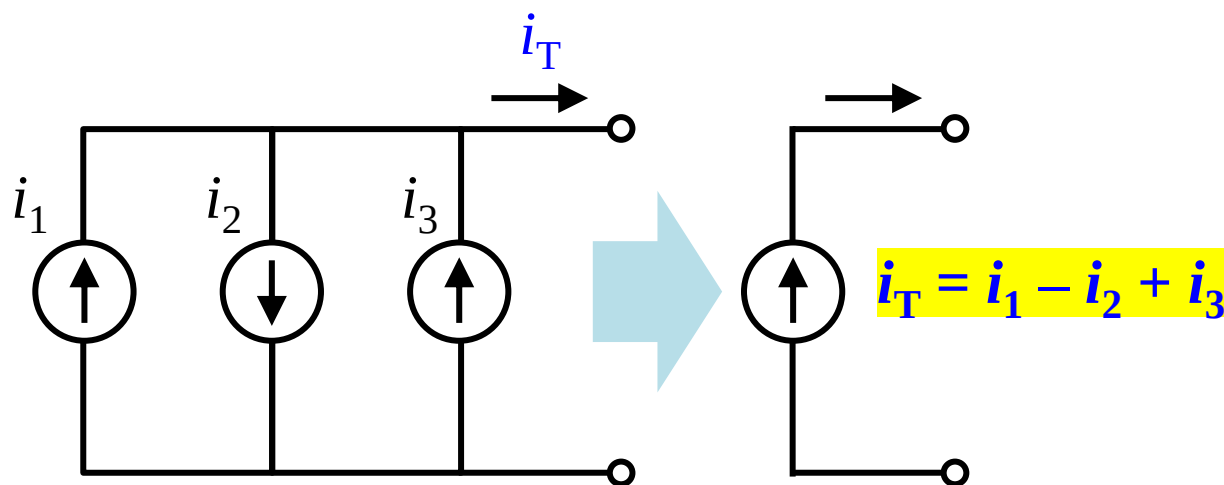
- a) At the port, the circuit's internal details, its design, and its theory of operation are no longer important. It becomes a building block.
- b) Equivalent circuits are equivalent as to external circuits, not internal.



2.7 Equivalent circuits (c)

- Equivalent current sources

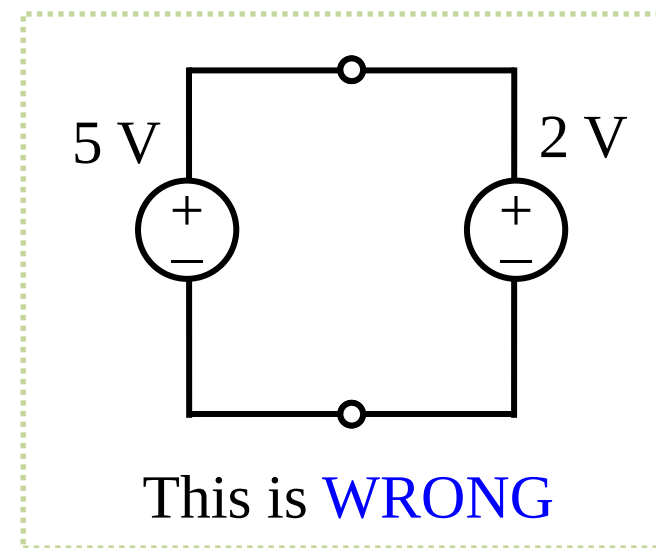
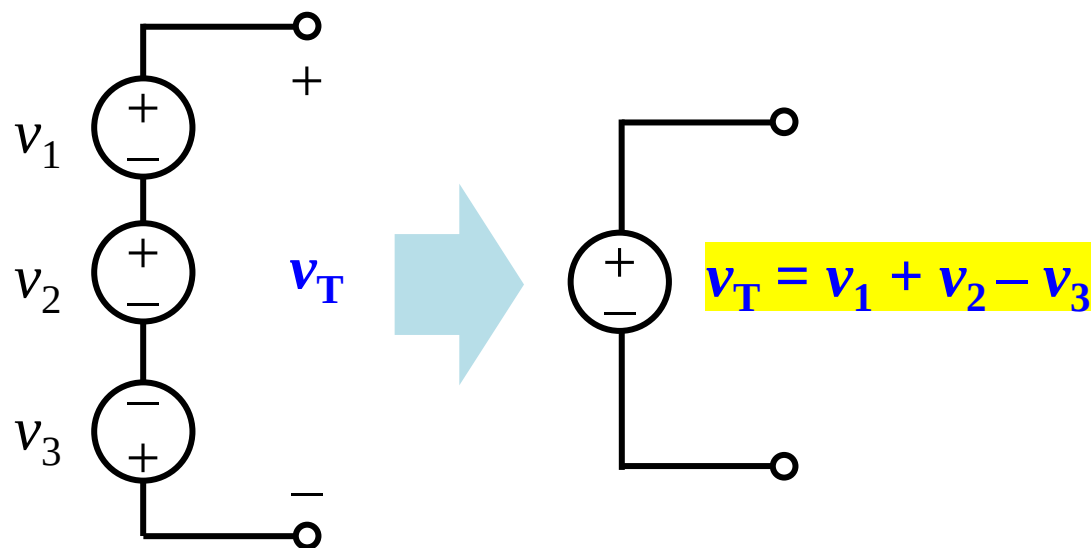
- a) When current sources are connected in parallel, the current sources could be summed up or combined with each other to create just one equivalent current source.
- b) Connecting current sources in series can be against KCL.



2.7 Equivalent circuits (d)

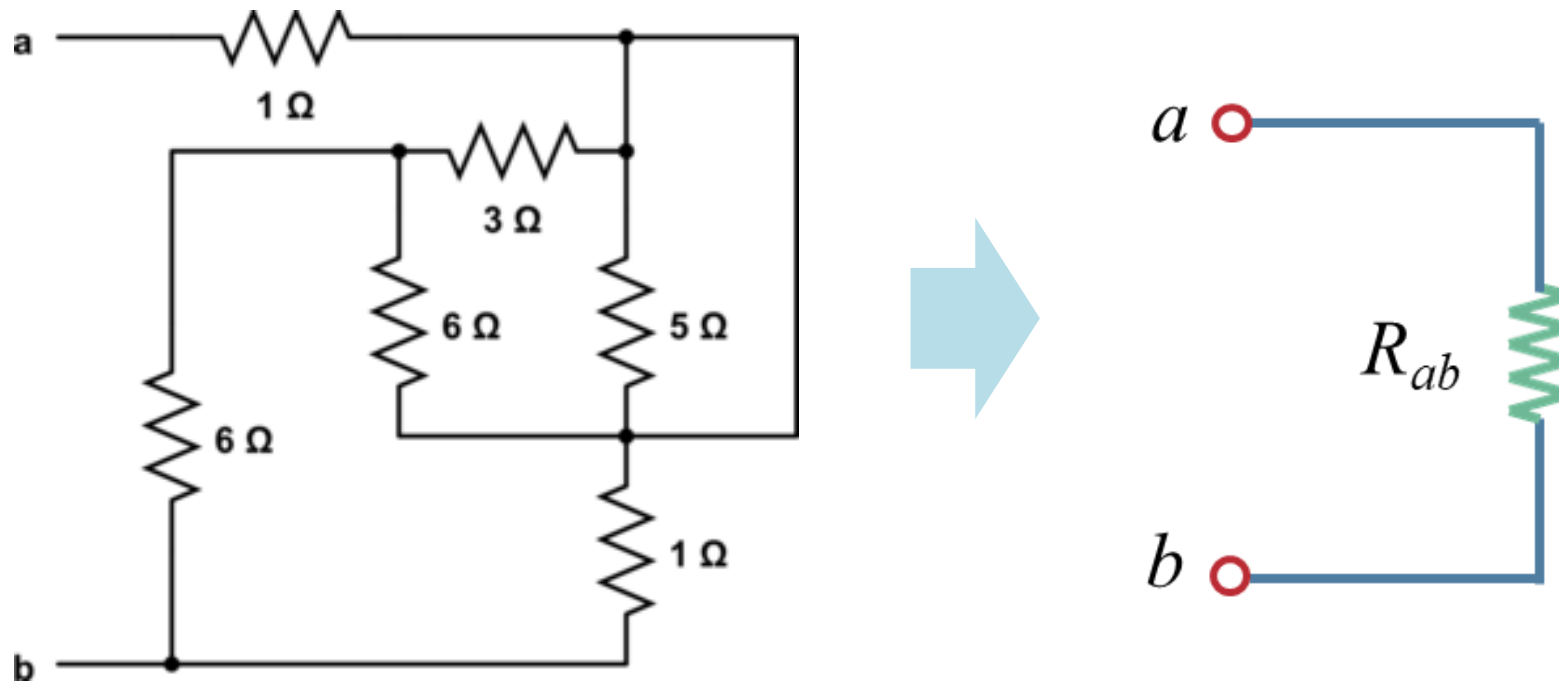
- Equivalent voltage sources

- When voltage sources are connected in series, the voltage sources could be summed up or combined with each other to create just one equivalent voltage source.
- Connecting voltage sources in parallel can be against KVL.



2.7 Equivalent circuits (e)

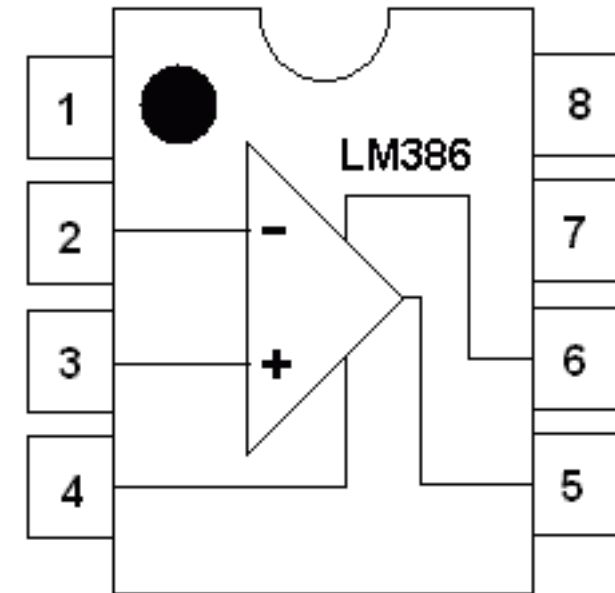
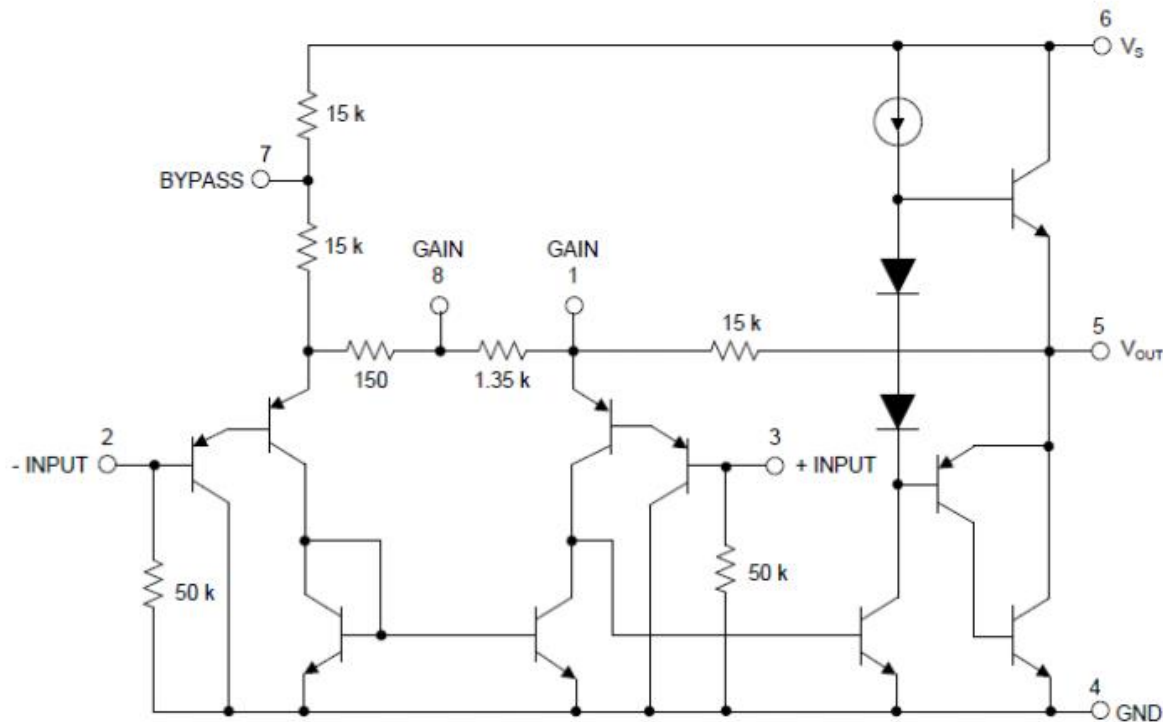
- Equivalent circuit of resistor networks



Is there a current passing through the $5\ \Omega$ resistor?

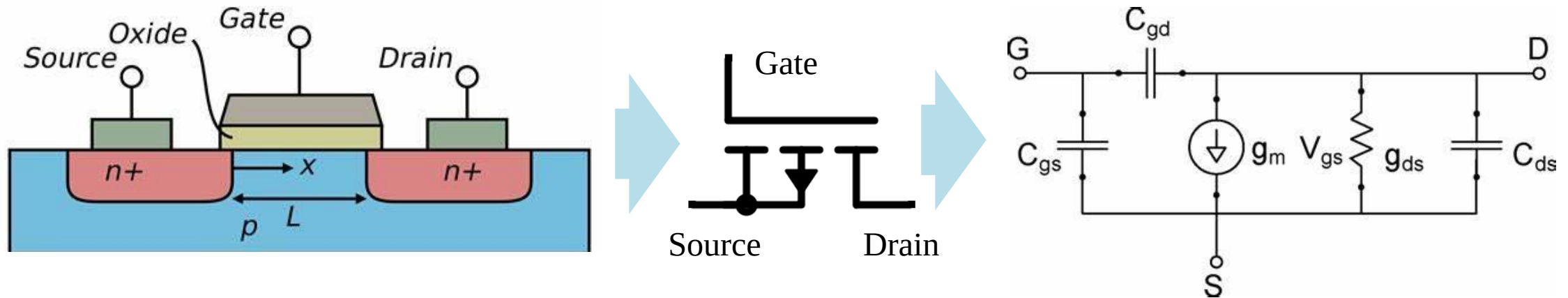
2.7 Equivalent circuits (f)

- Equivalent circuits of integrated circuits



We will revisit this concept in Lesson 5

Interesting To Know: Small-signal equivalent circuits of semiconductor devices



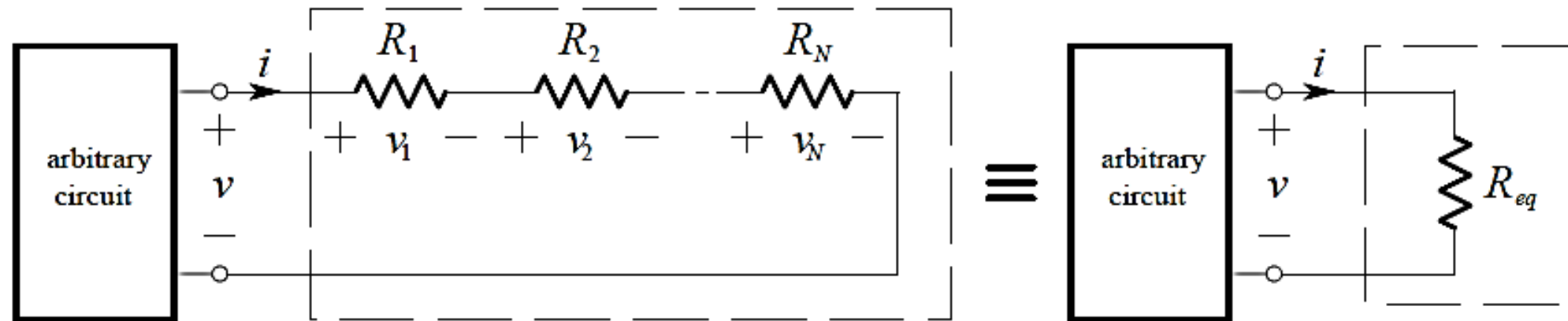
Small-signal modeling is a common analysis technique in electronics engineering which is used to approximate the behavior of electronic circuits containing nonlinear devices with linear equations. It is applicable to electronic circuits in which the AC signals, the time-varying currents and voltages in the circuit, have a small magnitude compared to the DC bias currents and voltages. A small-signal model is an AC equivalent circuit in which the nonlinear circuit elements are replaced by linear elements whose values are given by the first-order (linear) approximation of their characteristic curve near the bias point.

Chapter 2 Basic Laws

2.1-2.2	Ohm's law	Slides 09-20
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2.8 Series resistors (a)

- Series resistors



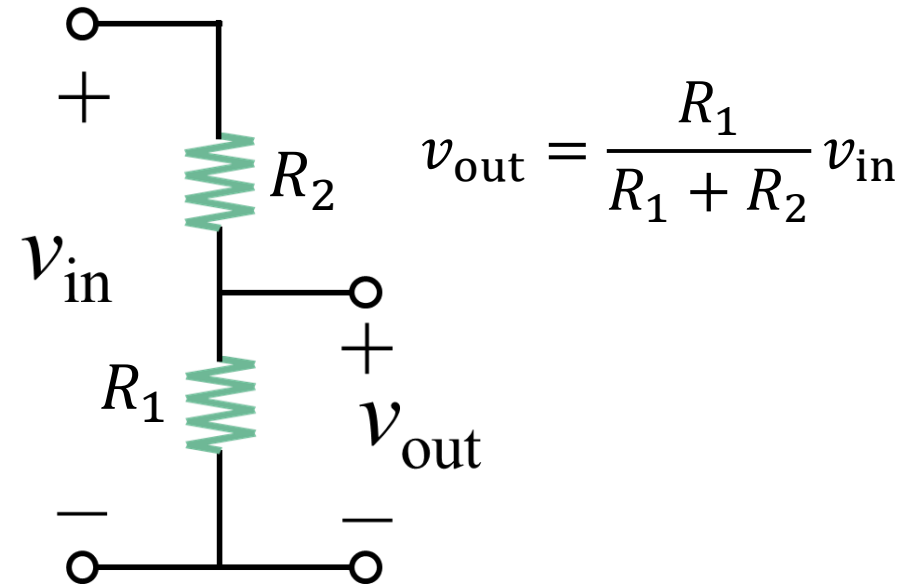
- a) Series resistors (串联): when resistors are in series, the current passing through them is the same.
- b) Equivalent resistance R_{eq} of series resistors is as follows:

$$R_{eq} = R_1 + R_2 + \dots + R_N = \sum_{n=1}^N R_N$$

2.8 Series resistors (b)

- Voltage dividers

- (a) A voltage divider (also known as a potential divider) is a passive linear circuit that produces an output voltage (V_{out}) that is a fraction of its input voltage (V_{in}) ($V_{\text{out}} < V_{\text{in}}$)
- (b) It can be based on resistors connected in series. Other devices like transistors can also be used to build voltage dividers.
- (c) It is often used in applications in which the required voltage may be variable.
- (d) v_{out} depends on the resistance of the load.



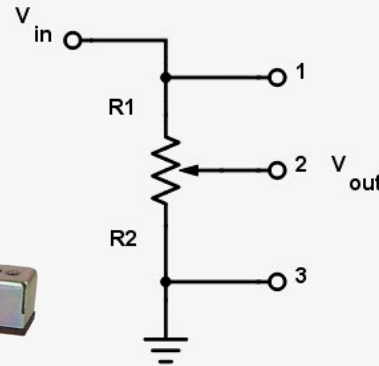
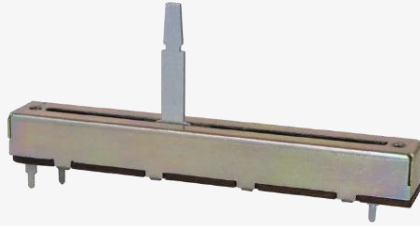
When multiple resistors are connected in series, the voltage across the n th resistor is :

$$v_n = \frac{R_n}{R_{\text{total}}} v_{\text{total}}$$

2.8 Series resistors (c)

- Application of voltage dividers (interesting to know)

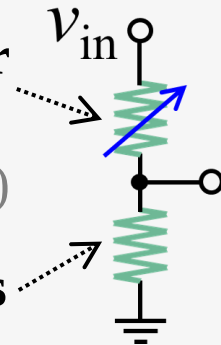
a) Potentiometer 分压计



b) Resistive sensors 电阻传感器

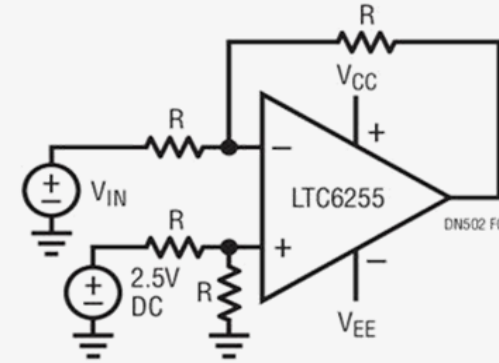
Variable resistor
(photocell, thermistor,
LDR, flex sensor, etc.)

Fixed resistors

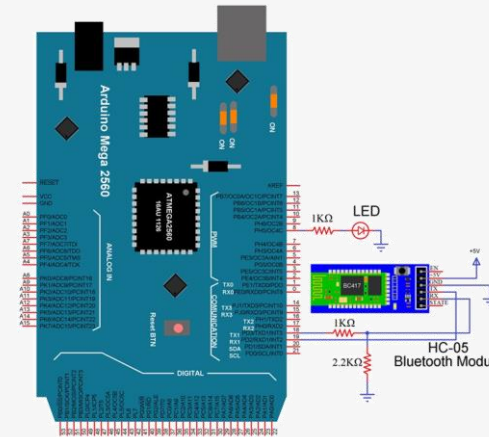


**V_{out} to microcontroller
analog input**

c) Level shifter 电平位移器



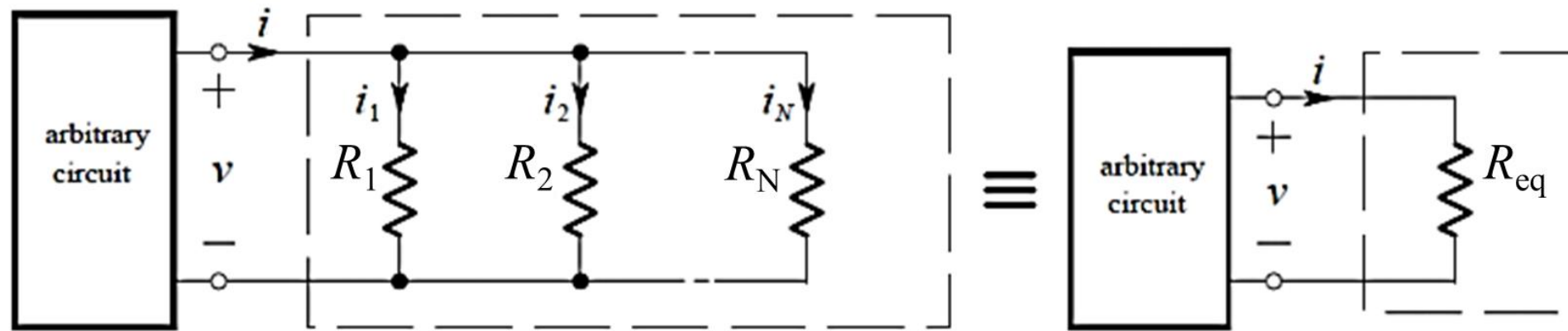
Often used in analog circuits like Op-Amp based circuits for example, where the required voltage may be variable.



Microcontrollers to Bluetooth, etc.
5 V to 3.3 V

2.9 Parallel resistors (a)

- Parallel resistors



- a) Parallel resistors (并联): when resistors are in parallel, the voltage drop across them is the same.
- b) Equivalent resistance R_{eq} of parallel resistors is as follows:

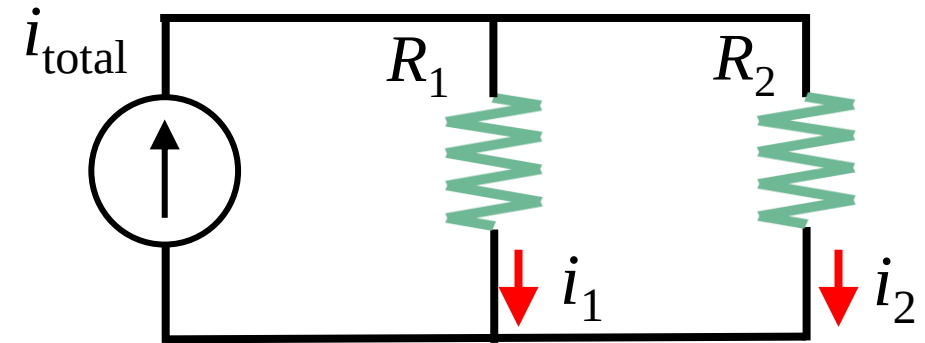
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N} \quad \text{Or} \quad G_{eq} = G_1 + G_2 + \dots + G_N = \sum_{n=1}^N G_N$$

2.9 Parallel resistors (b)

- Current dividers

- (a) A current divider is a linear circuit that produces an output current (i_{out}) that is a fraction of its input current (i_{in}). ($i_{\text{out}} < i_{\text{in}}$)
- (b) It can be based on resistors connected in parallel. Transistors, capacitors and inductors can also be used to build current dividers.
- (c) It is often used in current sensing, surge protection, etc.

$$i_1 = \frac{G_1}{G_1 + G_2} i_{\text{total}}$$



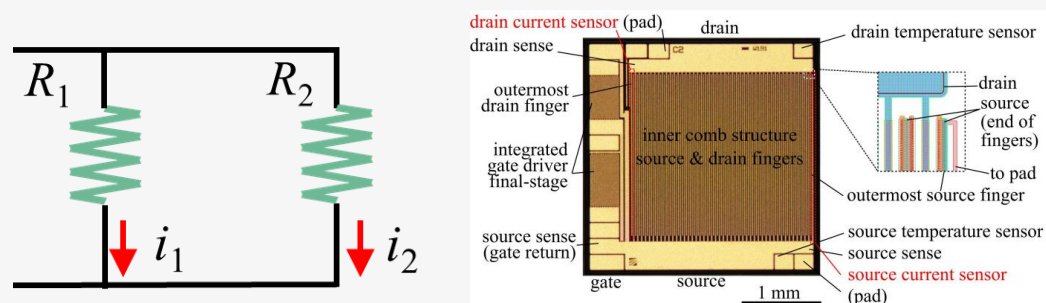
When multiple resistors are connected in parallel, the current through the n th resistor is :

$$i_n = \frac{G_n}{G_{\text{total}}} i_{\text{total}}$$

2.9 Parallel resistors (c)

- Application of current dividers (interesting to know)

a) Current monitoring 电流测量

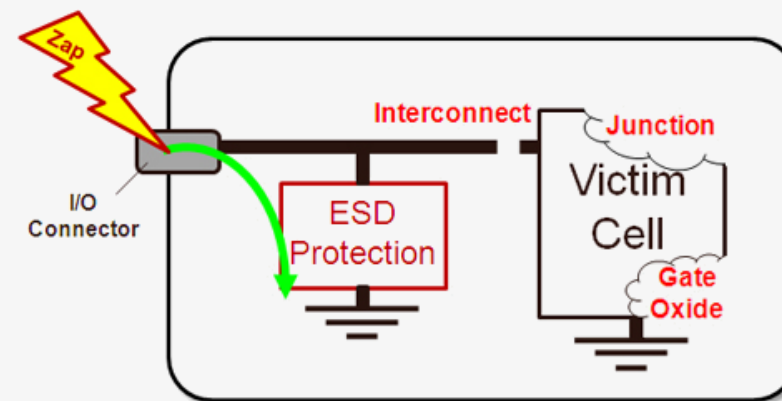
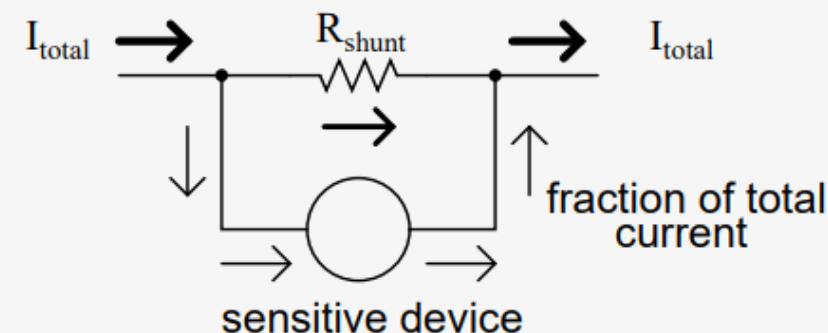


Power GaN ICs with integrated current sensing

We need to know i_2 yet it is highly inconvenient or inefficient to measure.

Then a resistor R_1 is put in parallel with R_2 , then i_1 can be measured and i_2 can be determined, which can be more convenient or efficient depending on the topologies.

b) ESD protection 静电保护



2.10 ∇ division and $/$ division rules (a)

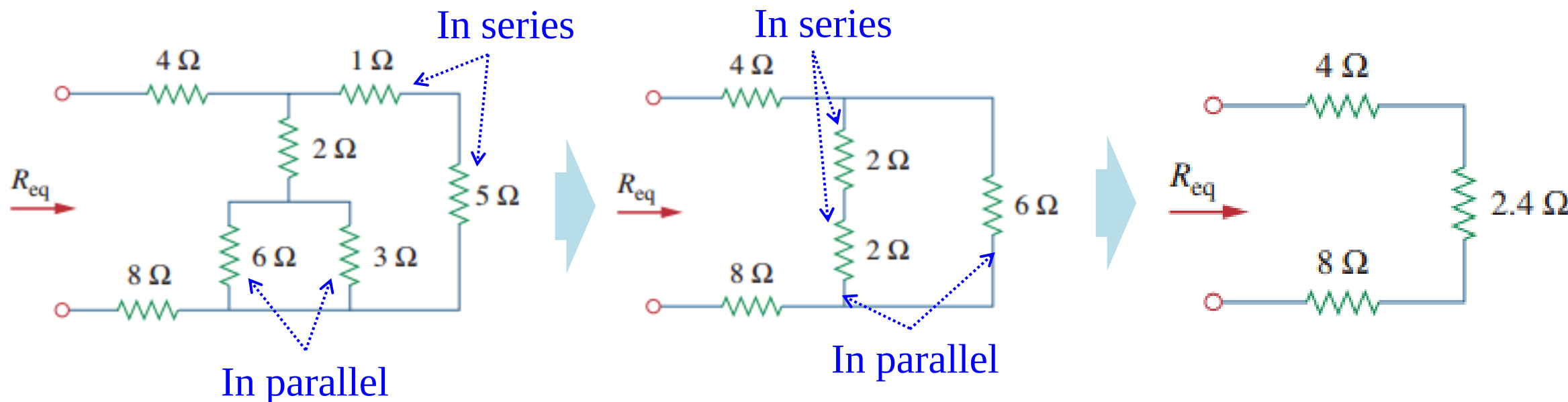
- Voltage and current dividers

- a) The division rules, when combined with the equivalent resistances, may be the most used technique in electronics, which will be shown in the following examples.
- b) Such methods allow us to quickly have ideas about what is going on in a circuit and about some important voltages and currents.
- c) An engineer could certainly analyze and design circuits without having voltage and dividers in their “tool bag”, but they would be wasting lots of time writing unnecessary KVL and KCL equations.

2.10 V division and / division rules (b)

- Example 07

Example 07: Find the R_{eq} in the following circuit.

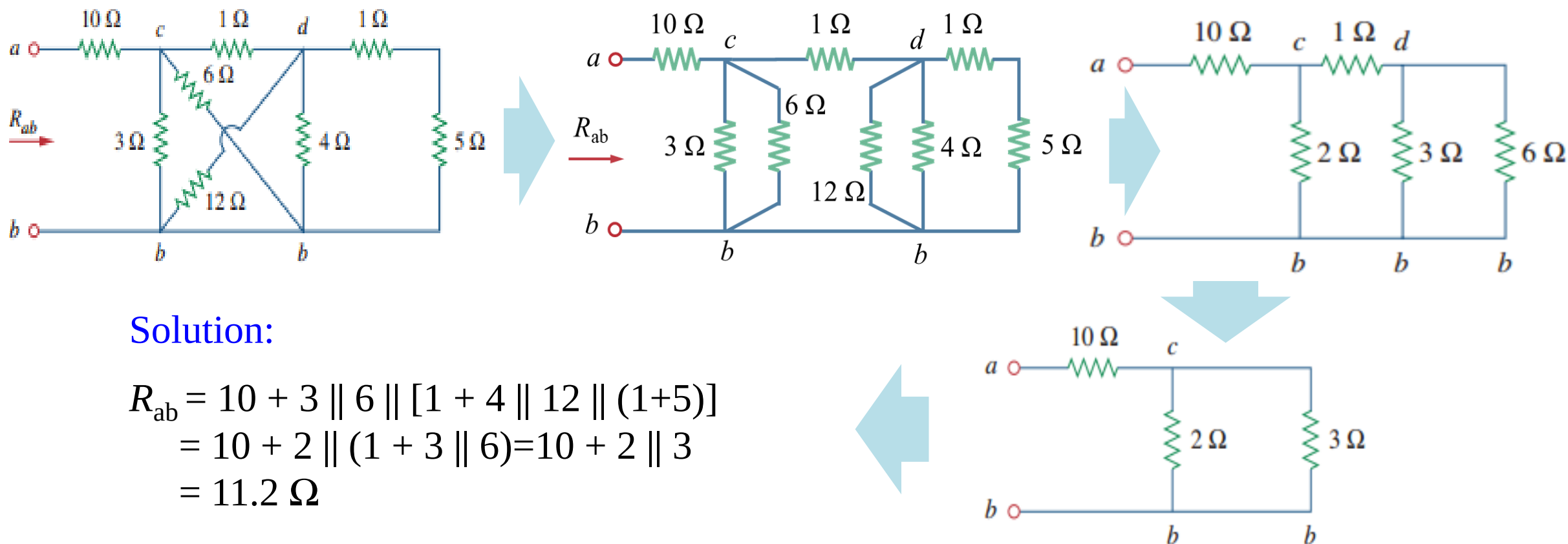


Solution: $R_{eq} = 4 + 8 + (1 + 5) \parallel [2 + (6 \parallel 3)] = 12 + 6 \parallel (2 + 2) = 12 + 6 \parallel 4 = 14.4 \Omega$

2.10 V division and / division rules (c)

- Example 08

Example 08: Calculate the equivalent resistance R_{ab} in the following circuit.



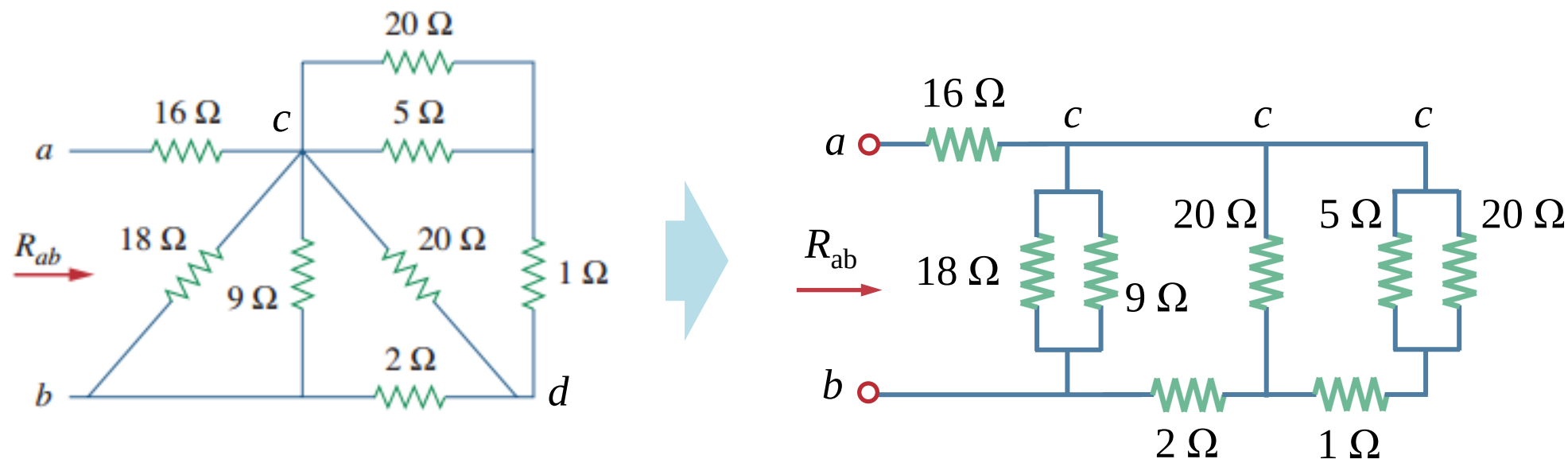
Solution:

$$\begin{aligned}
 R_{ab} &= 10 + 3 \parallel 6 \parallel [1 + 4 \parallel 12 \parallel (1+5)] \\
 &= 10 + 2 \parallel (1 + 3 \parallel 6) = 10 + 2 \parallel 3 \\
 &= 11.2 \Omega
 \end{aligned}$$

2.10 V division and / division rules (d)

- Example 09

Example 09: Calculate the equivalent resistance R_{eq} in the following circuit.

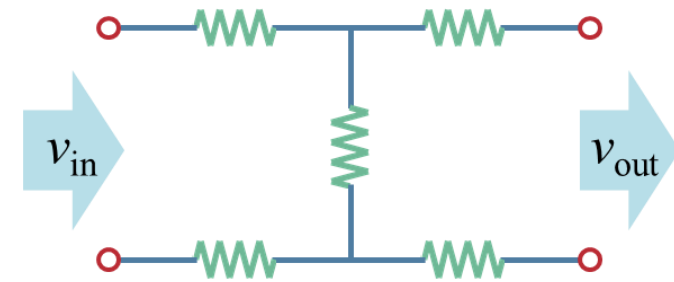


Solution: $R_{ab} = 16 + (18 \parallel 9) \parallel [20 \parallel (20 \parallel 5 + 1) + 2] = 19\ \Omega$

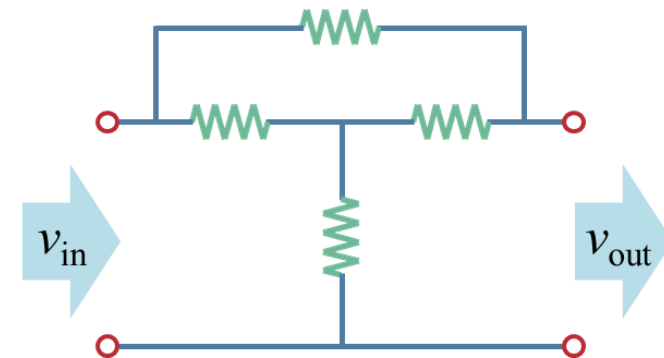
Interesting To Know: attenuators

- a) An attenuator is a two-port resistive network designed to weaken or “attenuate” the power being supplied by a source to a level that is suitable for the connected load.
- b) An attenuator is effectively the opposite of an amplifier, though they work by different methods. (We will talk about amplifiers in Lesson 5)
- c) Attenuators are usually made up of simple voltage divider networks in order to lower the voltage like a level shifter and also to improve impedance matching.

1. Balanced Resistor T Pad attenuator



2. Un-balanced bridged T pad attenuator



2.10 V/division and /division rules (e)

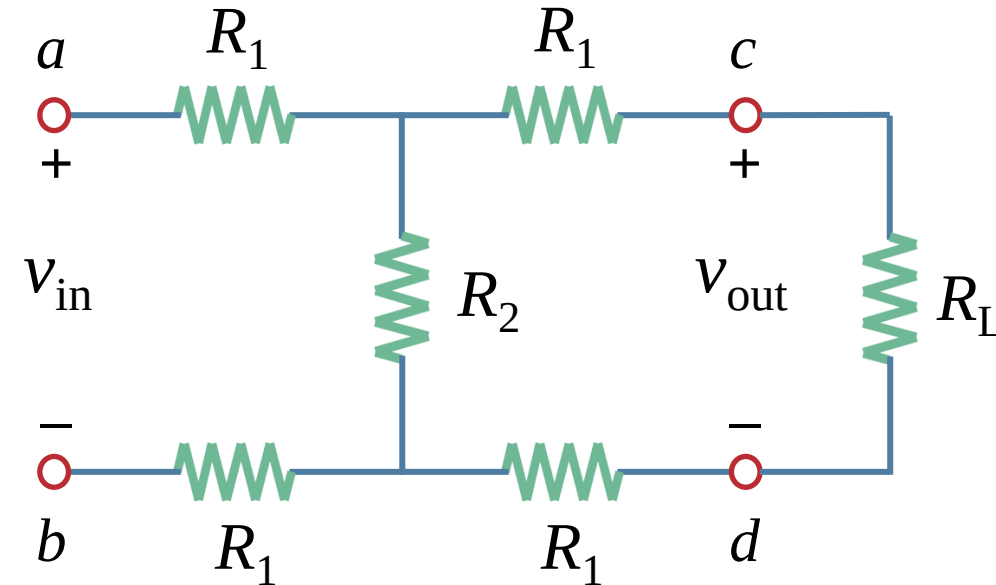
- Example 10

Example 10: The circuit in the right shows a typical fixed attenuator, connected to a load R_L . When designing an attenuator, the designer needs to select proper values for R_1 and R_2 , such that the $v_{\text{out}}/v_{\text{in}}$ ratio as well as the equivalent resistance between the terminals a and b are the desired ones.

(a) Prove if $R_{\text{ab}} = R_L$,

$$R_L^2 = 4R_1(R_1 + R_2) \quad \text{and} \quad \frac{v_{\text{out}}}{v_{\text{in}}} = \frac{R_2}{2R_1 + R_2 + R_L}$$

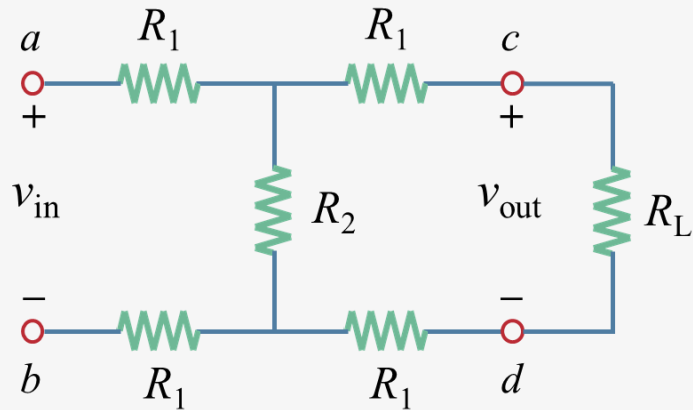
(b) Select Proper R_1 and R_2 , and make $R_{\text{ab}} = R_L = 300 \, \Omega$ and $v_{\text{out}}/v_{\text{in}} = 0.5$



2.10 V division and / division rules (e)

- Example 10

(a) Prove if $R_{ab} = R_L$, $R_L^2 = 4R_1(R_1 + R_2)$ and $\frac{v_{out}}{v_{in}} = \frac{R_2}{2R_1 + R_2 + R_L}$



Solution:

$$\begin{aligned}\text{Step 1: } R_L = R_{ab} &= 2R_1 + R_2 \parallel (2R_1 + R_L) \\ &= 2R_1 + R_2 (2R_1 + R_L) / (2R_1 + R_2 + R_L)\end{aligned}$$

$$R_L^2 = 4R_1^2 + 4R_1R_2 = 4R_1(R_1 + R_2)$$

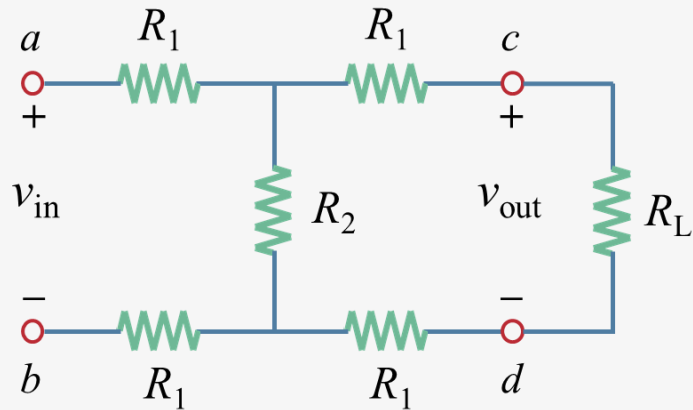
Step 2: using current division, the current in the R_L branch is:

$$i_L = \frac{v_{in}}{R_{ab}} \cdot \frac{\frac{1}{2R_1 + R_L}}{\frac{1}{R_2} + \frac{1}{2R_1 + R_L}} = \frac{v_{in}}{R_L} \cdot \frac{R_2}{2R_1 + R_2 + R_L} \Rightarrow v_0 = \frac{v_{in}}{R_L} \cdot \frac{R_2}{2R_1 + R_2 + R_L} \cdot R_L \Rightarrow \frac{v_{out}}{v_{in}} = \frac{R_2}{2R_1 + R_2 + R_L}$$

2.10 V division and I division rules (e)

- Example 10

(b) Select Proper R_1 and R_2 , and make $R_{ab} = R_L = 300\ \Omega$ and $v_{out}/v_{in} = 0.5$



Solution: it has been proved that when $R_{ab} = R_L$,

$$R_L^2 = 4R_1(R_1 + R_2) \quad \text{or} \quad \frac{v_{out}}{v_{in}} = \frac{R_2}{2R_1 + R_2 + R_L}$$

Now we need to know the R_1 and R_2 that makes $R_{ab} = R_L = 300\ \Omega$ and $v_{out}/v_{in} = 0.5$,

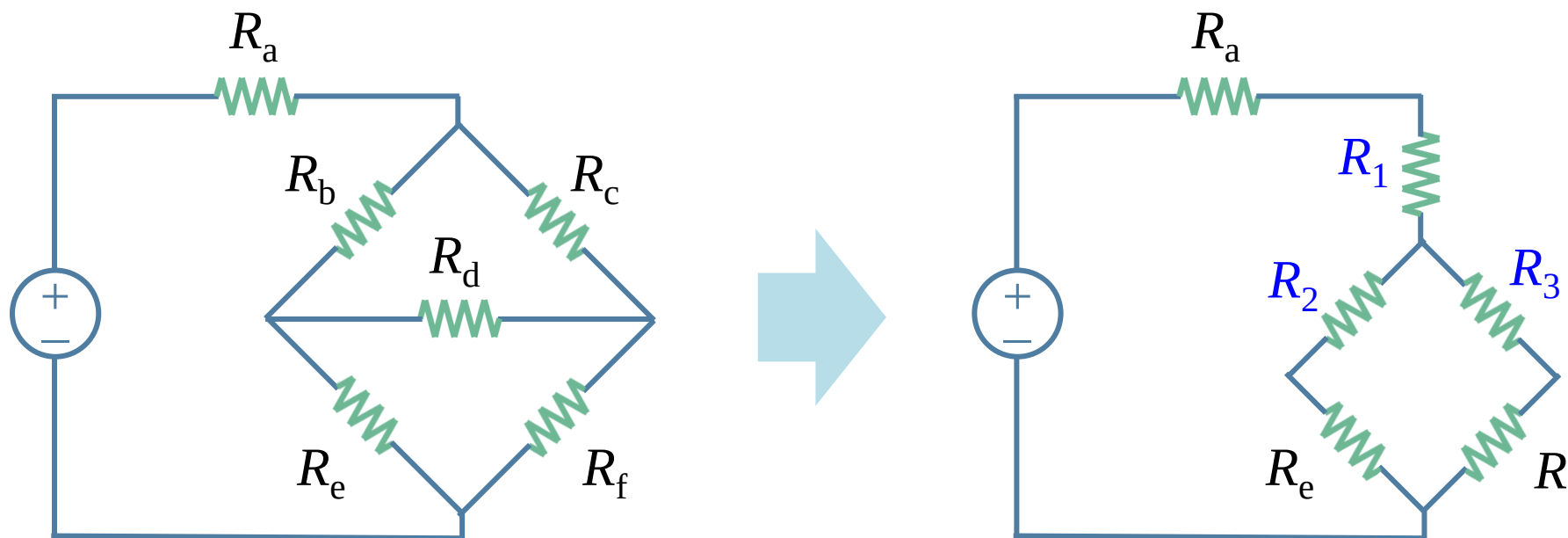
$$\left\{ \begin{array}{l} (300)^2 = 4(R_1 + R_2)R_1 \\ \frac{v_{out}}{v_{in}} = 0.5 = \frac{R_2}{2R_1 + R_2 + 300} \end{array} \right. \Rightarrow \begin{array}{l} R_1 = 50\ \Omega \\ R_2 = 400\ \Omega \end{array}$$

Chapter 2 Basic Laws

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2.11 Δ -Y and Y- Δ transformations (a)

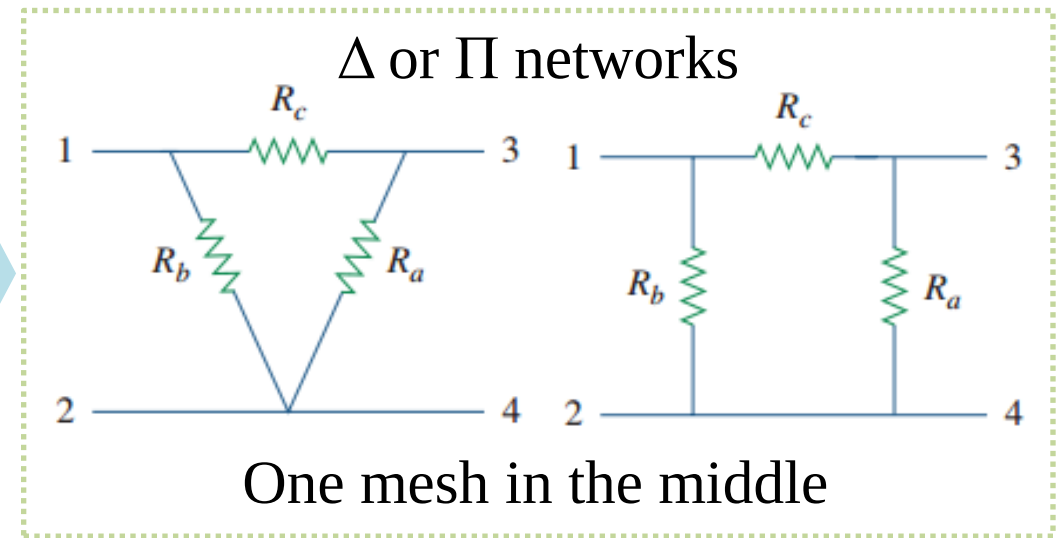
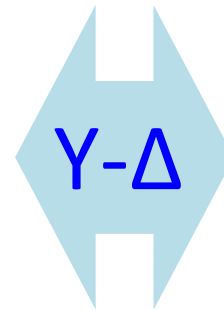
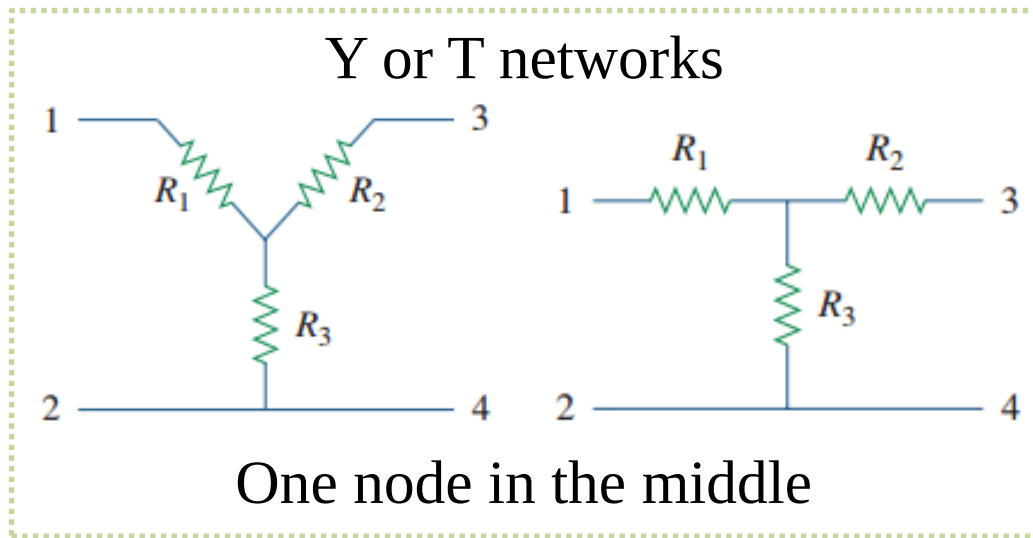
- Look at the circuit below, is it in parallel or in series?



- Topology transform (变换) allows for easy analysis of a circuit.
- To think about: will current pass through R_4 if $R_2 = R_3 = R_5 = R_6$, in the left circuit.

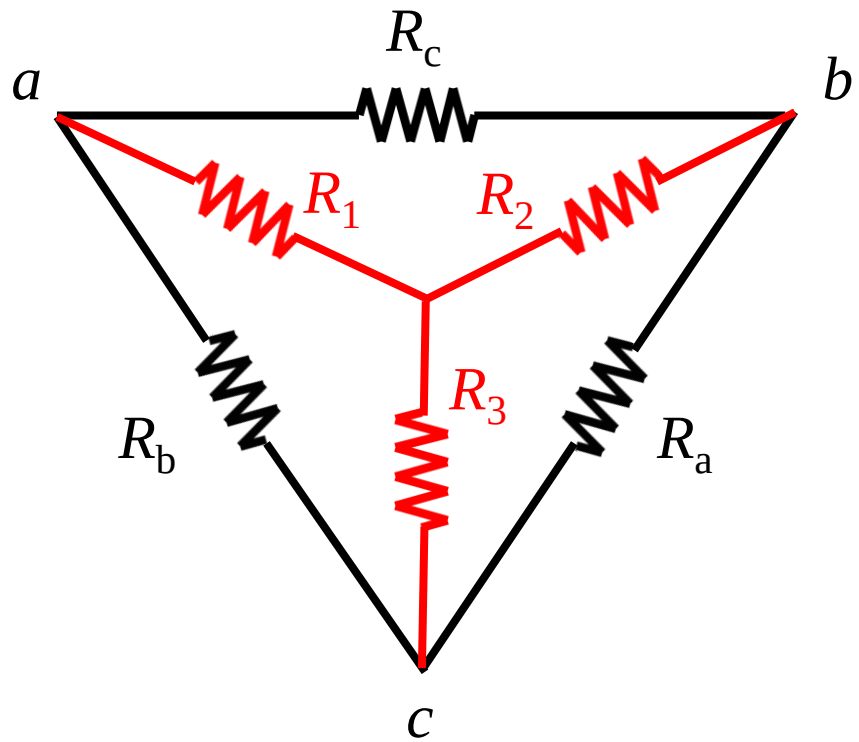
2.11 Δ -Y and Y- Δ transformations (b)

- Δ -Y and Y- Δ transformations
 - a) Many complex circuits can be simplified using series and/or parallel formulas, but sometimes circuit's element can't be categorized as series or parallel.
 - b) This transformation technique of the different circuitry was given by the Edwin Kennelly in 1899, who was an electrical engineer in the US.



2.11 Δ -Y and Y- Δ transformations (c)

- Δ -Y formulation



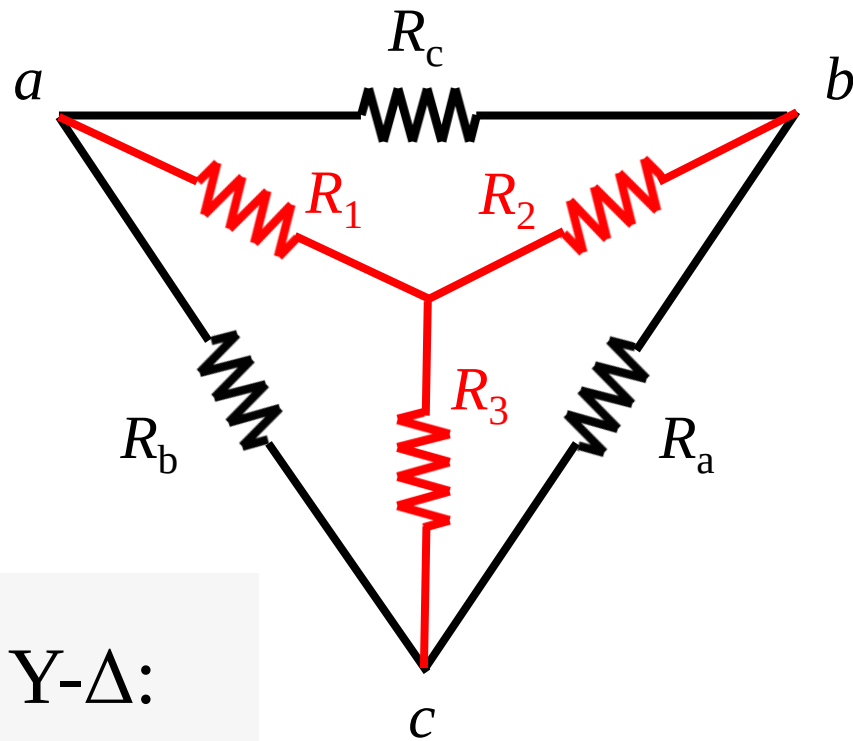
Ports	R_{eq}	$R_Y (Y)$	$R_\Delta (\Delta)$
a-c	$R_{ac} = R_1 + R_3 = R_b \parallel (R_c + R_a) = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c}$		

Δ -Y

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}, \quad R_2 = \frac{R_a R_c}{R_a + R_b + R_c}, \quad R_3 = \frac{R_b R_a}{R_a + R_b + R_c}$$

2.11 Δ -Y and Y- Δ transformations (d)

- Y- Δ formulation



Y- Δ :

Ports	R_{eq}	R_Y (Y)	R_Δ (Δ)
a-c	$R_{ac} = R_1 + R_3 = R_b \parallel (R_c + R_a) = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c}$		
a-b	$R_{ab} = R_1 + R_2 = R_c \parallel (R_a + R_b) = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c}$		
b-c	$R_{bc} = R_2 + R_3 = R_a \parallel (R_b + R_c) = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c}$		

$$R_a = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1}, \quad R_b = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2}, \quad R_c = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_3}$$

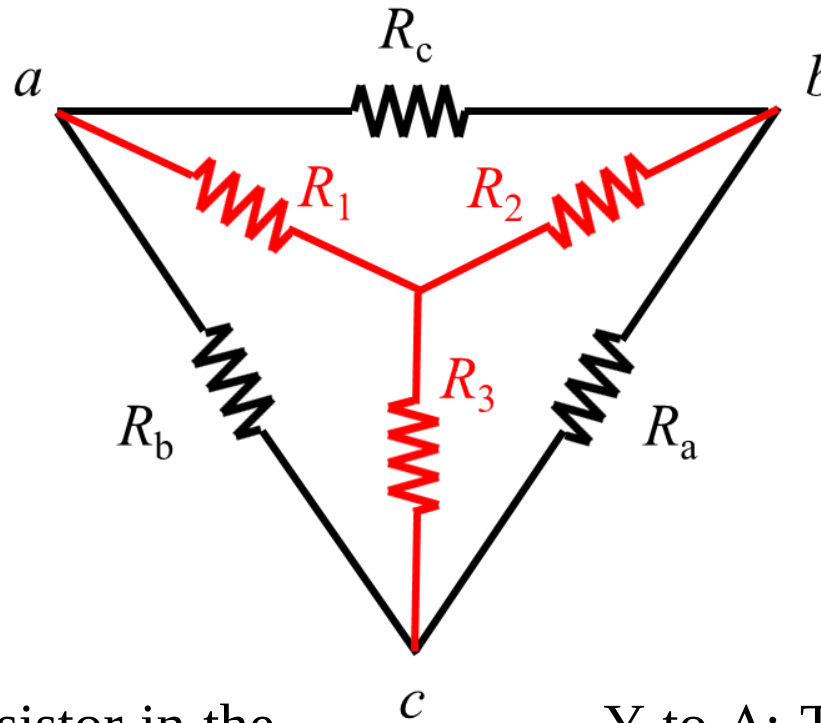
2.11 Δ -Y and Y- Δ transformations (e)

Δ to **Y**

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_b R_a}{R_a + R_b + R_c}$$



Y to Δ

$$R_a = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2}$$

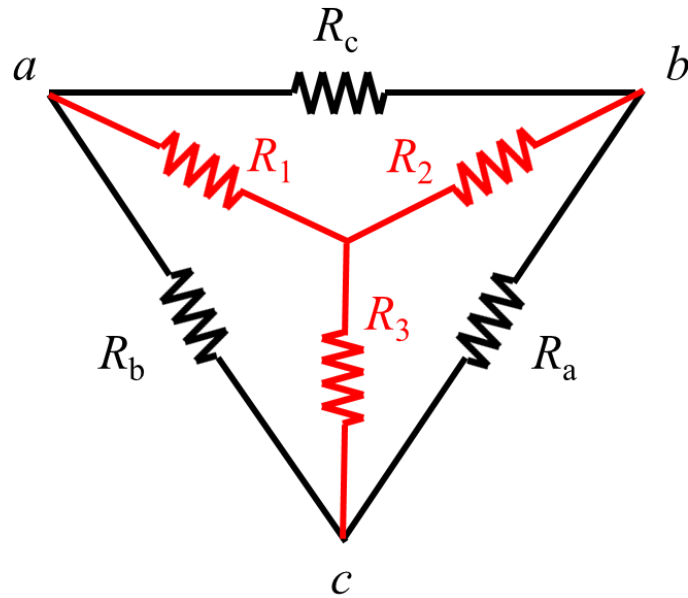
$$R_c = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_3}$$

Δ to Y: the resistance of a resistor in the Y network is the product of the resistance of the two adjacent resistors in the Δ network, divided by the sum of the resistance of the three Δ resistors.

Y to Δ : The resistance of a resistor in the Δ network is the sum of all possible products the resistance of the Y resistors taken two at a time, divided by the resistance of the opposite Y resistor.

2.11 Δ -Y and Y- Δ transformations (f)

- Comparison of R_Δ and R_Y



a) R_Y is less than R_Δ (外大内小), because a Δ network is more like a parallel connection, while a Y network is more like a series connection.

b) The Y and Δ networks are said to be balanced when,

$$R_a = R_b = R_c = R_\Delta \quad R_1 = R_2 = R_3 = R_Y$$

In such case we have:

$$R_Y = R_\Delta / 3 \quad \text{or} \quad R_\Delta = 3R_Y$$

Note that in making the transformation, we do not take anything out of the circuit or put in anything new.

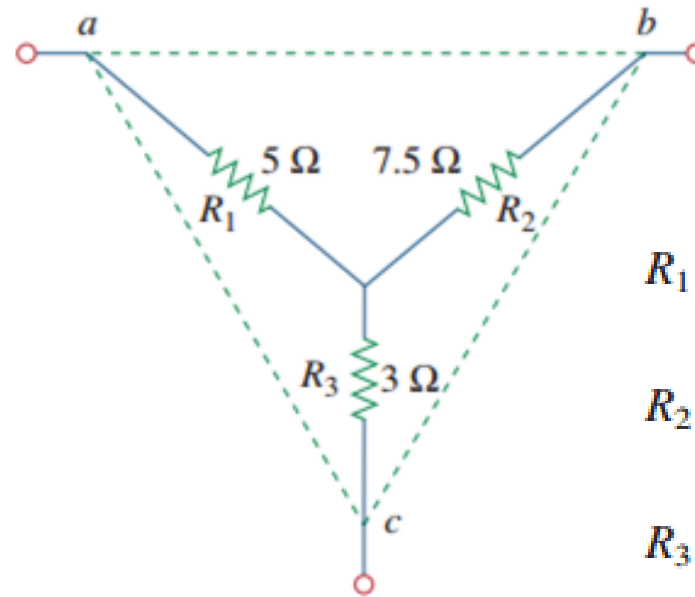
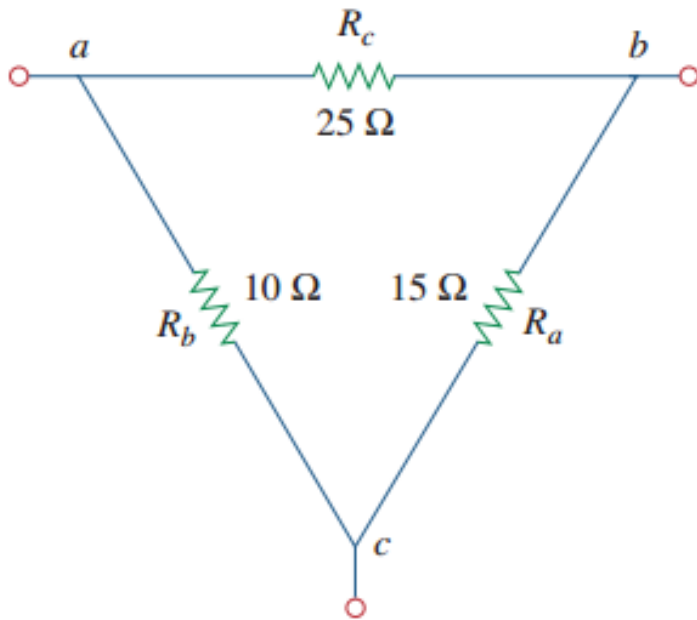
We are merely substituting different but mathematically equivalent three-terminal network patterns to create a circuit in which resistors are either in series or in parallel, allowing us to calculate R_{eq} if necessary.

2.11 Δ -Y and Y- Δ transformations (g)

- Example 11

Example 11: Convert the Δ network in to an equivalent Y network.

Solution:



$$R_1 = \frac{R_a R_b}{R_a + R_b + R_c}$$

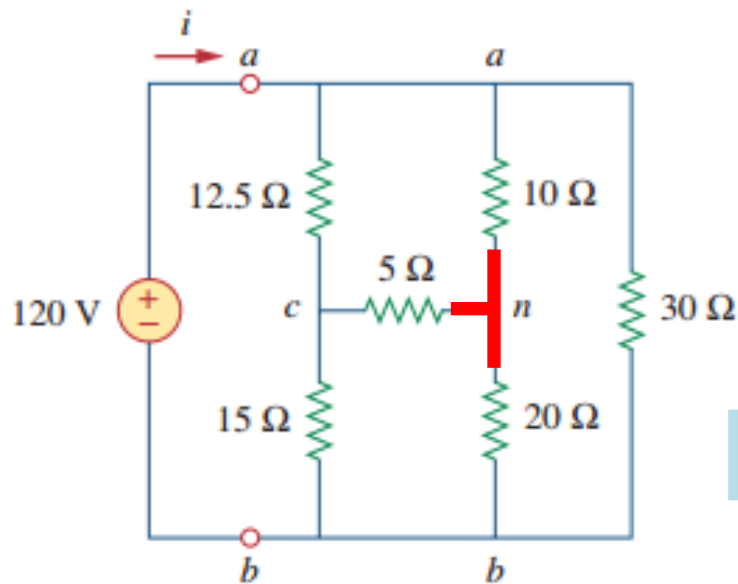
$$R_2 = \frac{R_a R_b}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

2.11 Δ -Y and Y- Δ transformations (h)

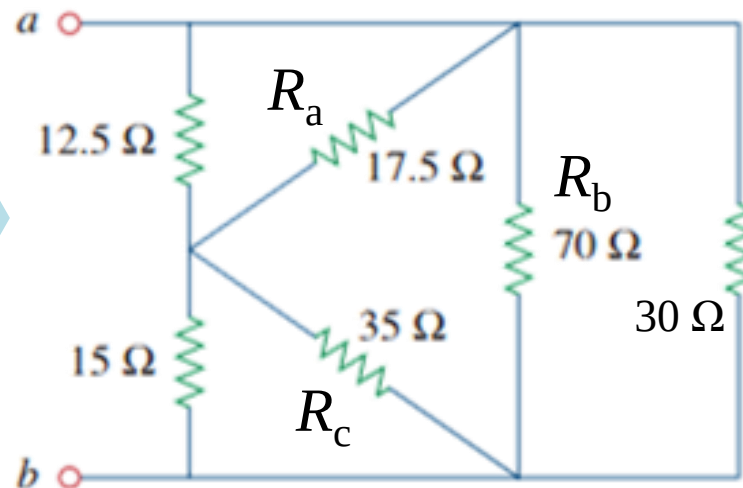
- Example 12

Example 12: Obtain the equivalent R_{ab} for the circuit shown below and use it to find i .



Solution I (Y- Δ):

Step 1. Use Y- Δ transformation to simplify the circuit



$$R_a = \frac{10 \times 20 + 20 \times 5 + 5 \times 10}{30}$$

$$R_b = \frac{10 \times 20 + 20 \times 5 + 5 \times 10}{30}$$

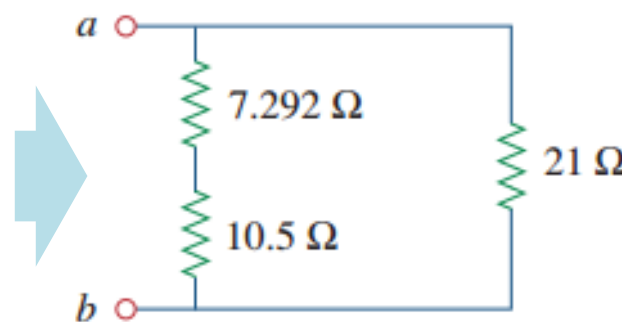
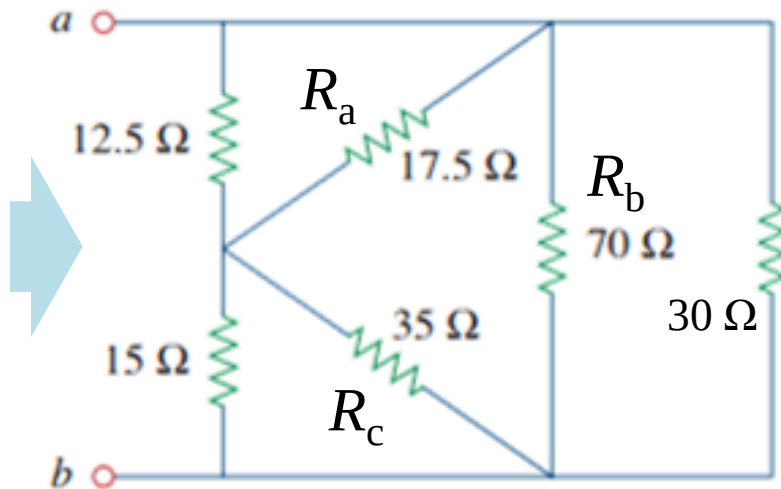
$$R_c = \frac{10 \times 20 + 20 \times 5 + 5 \times 10}{30}$$

2.11 Δ -Y and Y- Δ transformations (i)

- Example 12

Example 12: Obtain the equivalent R_{ab} for the circuit shown below and use it to find i .

Solution I (Y- Δ): Step 2. Use voltage division and current division rules to simplify the circuit and calculate R_{ab} .



$$12.5 \parallel 17.5 = \frac{12.5 \times 17.5}{12.5 + 17.5} = 7.292 \Omega$$

$$15 \parallel 35 = \frac{15 \times 35}{15 + 35} = 10.5 \Omega \quad 70 \parallel 30 = \frac{70 \times 30}{70 + 30} = 21 \Omega$$

$$R_{ab} = (7.292 + 10.5) \parallel 21 = \frac{17.792 \times 21}{17.792 + 21} = 9.632 \Omega$$

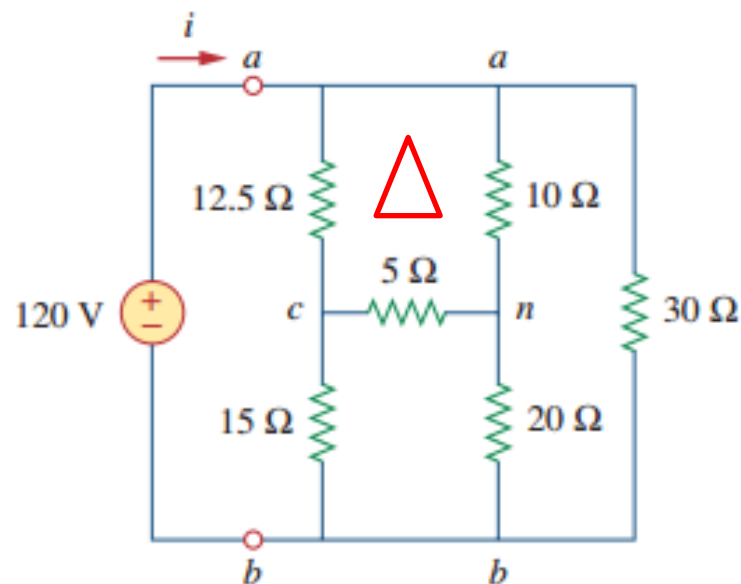
Step 3. Use Ohm's Law to calculate the current.

$$i = \frac{v_s}{R_{ab}} = \frac{120}{9.632} = 12.458 \text{ A}$$

2.11 Δ -Y and Y- Δ transformations (j)

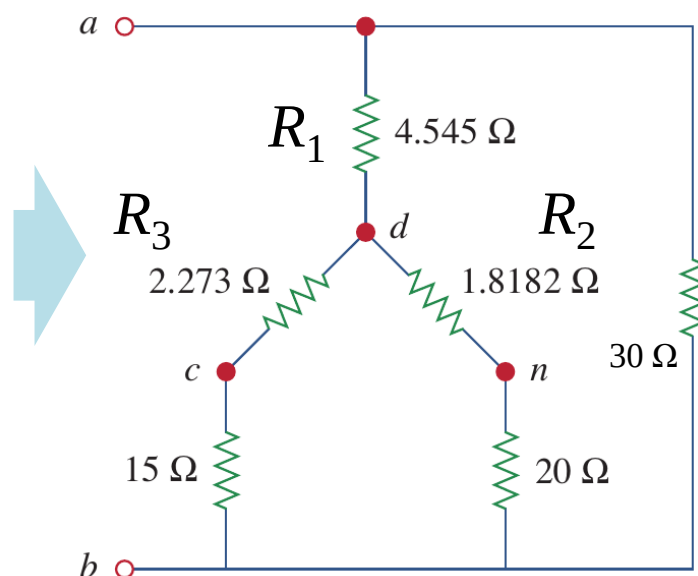
- Example 12

Example 12: Obtain the equivalent R_{ab} for the circuit shown below and use it to find i .



Solution II (Δ -Y):

Step 1. Use Δ -Y transformation to simplify the circuit



$$R_1 = \frac{12.5 \times 10}{12.5 + 10 + 5} = 4.545\Omega$$

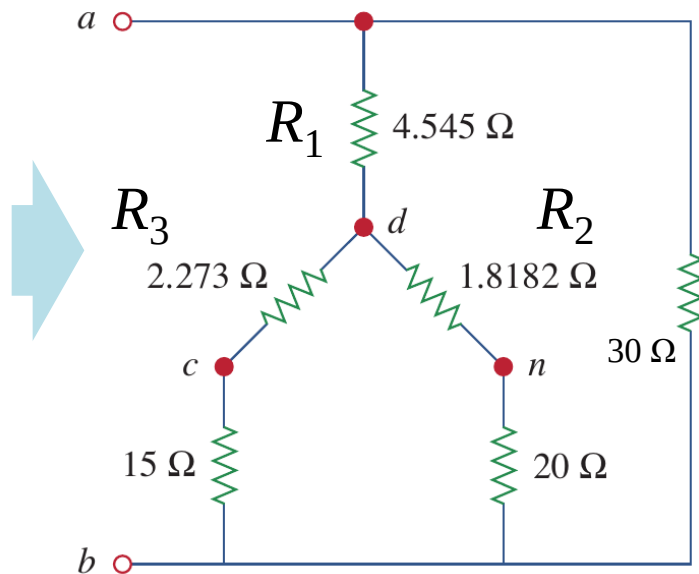
$$R_2 = \frac{5 \times 10}{12.5 + 10 + 5} = 1.8182\Omega$$

$$R_3 = \frac{5 \times 12.5}{12.5 + 10 + 5} = 2.273\Omega$$

2.11 Δ -Y and Y- Δ transformations (k)

- Example 12

Example 12: Obtain the equivalent R_{ab} for the circuit shown below and use it to find i .



Solution II (Δ -Y):

Step 2. Use voltage division and current division rules to simplify the circuit and calculate R_{ab} .

$$R_{ab} = [4.545 + (2.273 + 15) \parallel (1.8182 + 20)] \parallel 30 = 9.631 \Omega$$

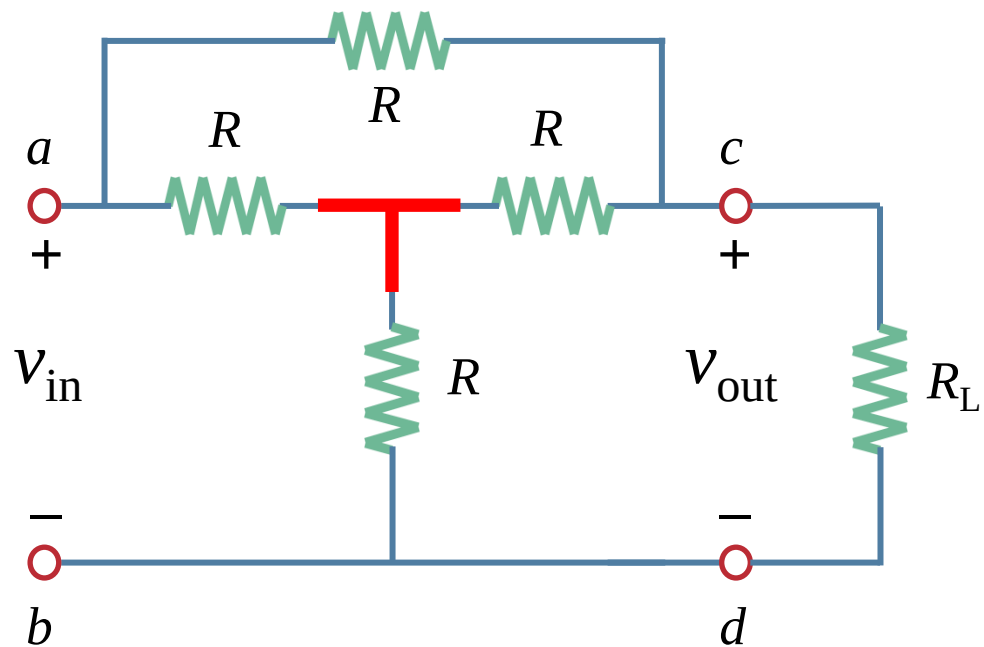
Step 3. Use voltage division and current division rule to simplify the circuit.

$$i = v_s / R_{ab} = 12.46 \text{ A}$$

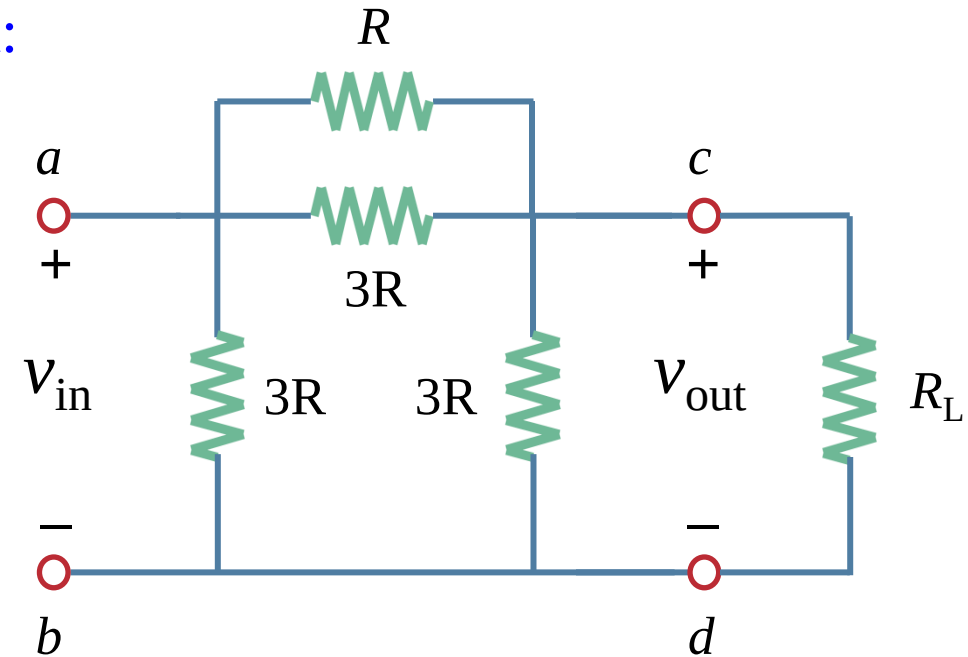
2.11 Δ -Y and Y- Δ transformations (I)

- Example 13

Example 13: The fixed attenuator shown in the right figure is called T bridge. Prove, if $R = R_L$, $R_{ab} = R_L$ and $v_{out}/v_{in} = 0.5$.



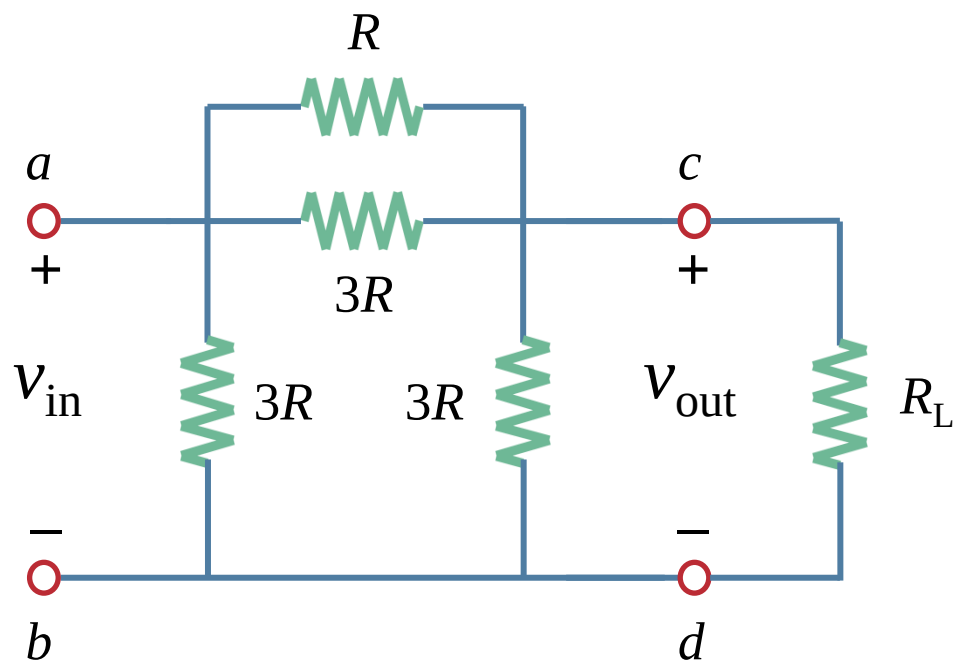
Solution:



2.11 Δ -Y and Y- Δ transformations (m)

- Example 13

Example 13: The fixed attenuator shown in the right figure is called T bridge. Prove, if $R = R_L$, $R_{ab} = R_L$ and $v_{out}/v_{in} = 0.5$.



Solution:

$$\begin{aligned} R_{ab} &= 3R \parallel (R \parallel 3R + R_L \parallel 3R) = 3R \parallel (0.75R + 0.75R) \\ &= 3R \parallel 1.5R = R \end{aligned}$$

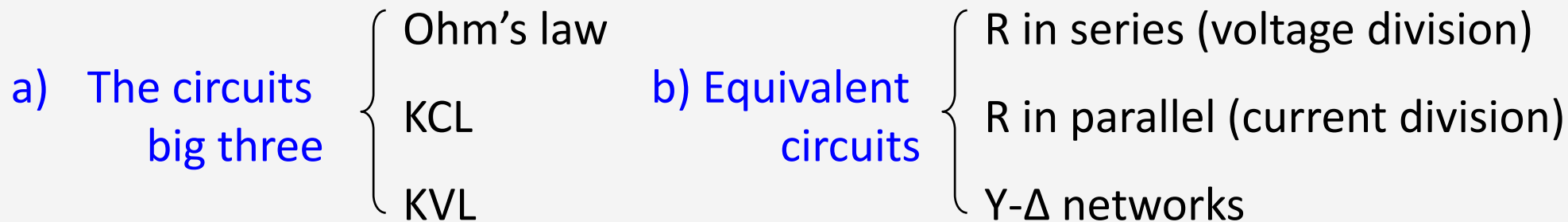
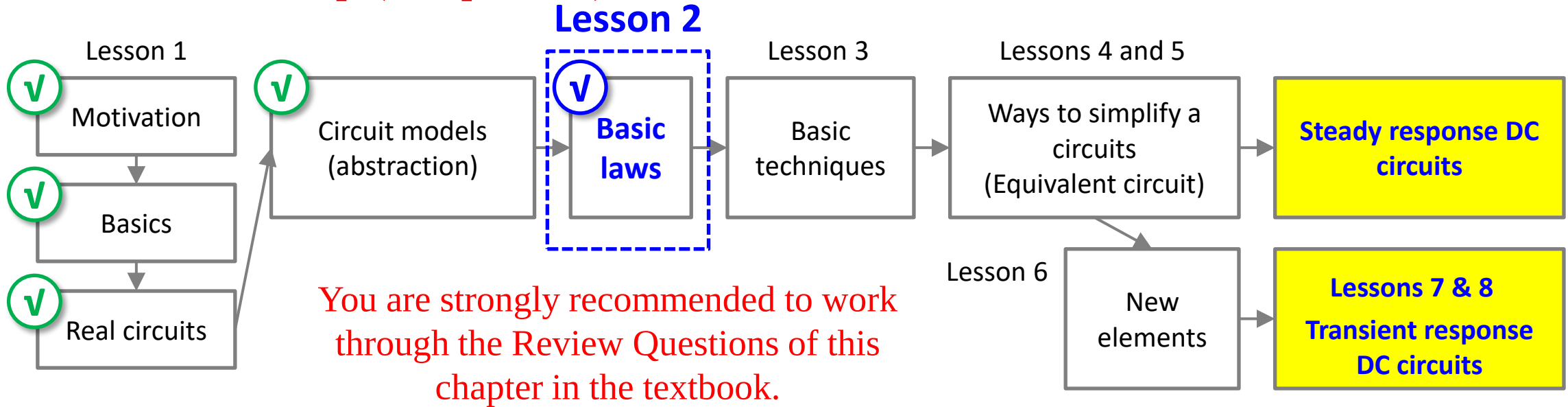
$$v_{out} = v_{in} \cdot \frac{R_L \parallel 3R}{R \parallel 3R + R_L \parallel 3R} = v_{in} \cdot 0.5 \Rightarrow \frac{v_{in}}{v_{out}} = 0.5$$

Chapter 2 Basic Laws

2.1-2.2	Ohm's law	Slides 09-20
2.3-2.6	Kirchhoff's Current & voltage Laws (KCL & KVL)	Slides 22-39
2.7	Equivalent circuits	Slides 41-47
2.8-2.10	Series and parallel resistors	Slides 49-62
2.11	Δ -Y and Y- Δ transformations	Slides 64-76
2.12	Summary	Slides 78-79
2.13	Assignments	Slide 80

2.12 Summary (a)

- EE104 roadmap (DC portion)



2.12 Summary (b)

- Important updates

Name	Class	Duty	TA office hour	TA office	E-mail
李熠 Yi, LI	1	assignments exams QA	Thursday 19:00-20:00	Rm 1001, Nanshan Intelligent Park	1029879532@qq.com
肖祺祺 Qiqi, XIAO	1	assignments exams QA	Thursday 15:00-16:00	Rm 525, Northern Tower Engineering building	12433012@mail.sustech. edu.cn
庄泽咏 Zeyong, ZHUANG	4	assignments exams QA	Monday 16:00-17:00	Rm G305, LYNN LIBRARY	1270881472@qq.com
洪朴 Pu, HONG	4	assignments exams QA	Wednesday 15:00-16:00	Rm 230, Southern Tower, Engineering building	12332195@mail.sustech. edu.cn

2.13 Assignments

- To be submitted on March 4th (class 1) and 5th (class 2)

Chapter 2:

Problems 2.6, 2.13, 2.14, 2.25, 2.57

Write down or print your answers and submit during the class.

Remember to put your name, student ID, and class number.

There are far more problems in the textbook, try to complete as many as you can.
They will not take you too much time if you know the tricks.

Shelf clouds

